

The novel role of Kallistatin in linking metabolic syndromes and cognitive memory deterioration by inducing amyloid- β plaques accumulation and tau protein hyperphosphorylation

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Abstract

Accumulation of amyloid β (A β) peptides and hyperphosphorylated tau proteins in the hippocampus triggers cognitive memory decline in Alzheimer's disease (AD). The incidence and mortality of sporadic AD were tightly associated with diabetes and hyperlipidemia, while the exact linked molecular is uncertain. Here, we reported that serum Kallistatin concentrations were meaningfully higher in AD patients, with a higher concentration of fasting blood glucose and triglyceride. In addition, the constructed Kallistatin-transgenic (KAL-TG) mice defined its cognitive memory impairment phenotype and lower LTP in hippocampal CA1 neurons accompanied by increased A β deposition and tau phosphorylation. Mechanistically, Kallistatin could directly bind to the Notch1 receptor and thereby upregulate BACE1 expression by inhibiting PPAR γ signaling, resulting in A β cleavage and production. Besides, Kallistatin could promote the phosphorylation of tau by activating GSK-3 β . Fenofibrate, a hypolipidemic drug, could alleviate cognitive memory impairment by down-regulating A β and tau phosphorylation of KAL-TG mice. Collectively, our data clarified a novel mechanism for A β accumulation and tau protein hyperphosphorylation regulation by Kallistatin, which might play a crucial role in linking metabolic syndromes and cognitive memory deterioration, and suggested that fenofibrate might have the potential for treating metabolism-related AD.

Highlights

- Kallistatin-transgenic(KAL-TG) mice defined its cognitive memory impairment phenotype accompanied by increased A β deposition and tau phosphorylation.
- Kallistatin could directly bind to the Notch1 receptor and thereby upregulate BACE1 expression by inhibiting PPAR γ signaling.
- Fenofibrate could alleviate cognitive memory impairment and down-regulate the serum Kallistatin level.

eLife assessment

This study identified an innovative molecular mechanism linking diabetes to Alzheimer's disease (AD) risk, with **important** significance. The finding presents novel insights into AD pathogenesis and provides strong evidence about the mechanistic roles of Kallistatin, and the therapeutic potential of fenofibrate in AD. The experiments are well conducted, and the evidence is **convincing**.

<https://doi.org/10.7554/eLife.99462.1.sa3>

Introduction

Alzheimer's disease (AD), the most common irreversible neurodegenerative disease involving dementia in the elderly, is characterized by progressive deterioration of cognitive memory. Pathologically, the characteristic hallmarks of AD are extracellular amyloid- β (A β) plaques and intracellular neurofibrillary tangles (NFTs) ^{1,2,3}. A β cascade and the hyperphosphorylation of tau protein are two main hypotheses related to AD. The A β cascade hypothesis dominates that overproduction of A β changes the normal cellular state, resulting in synaptic dysfunction, neurodegeneration, tau hyperphosphorylation, and neuroinflammation, eventually causing memory loss in AD and dementia patients ^{4,5}. A β peptides are produced from the sequential cleavage of amyloid precursor protein (APP) by β -secretase (β -site APP cleaving enzyme 1, BACE1) and γ -secretase, therefore this cleavage step contributes heavily to AD pathology ^{6,7}. BACE1 is one of the most potential therapeutic targets. Some potent BACE1 inhibitors have advanced to late stages in clinical trials, highlighting the significance of BACE1 in the A β generation ^{8,9,10}. Tau is a protein found in microtubules. It exists physiologically in axons and regulates microtubule dynamics and axonal transport ¹¹. Tau goes through a multi-step process in AD, transitioning from a natively unfolded monomer to massive aggregated aggregates like NFTs, another hallmark of AD ^{12,13}. One of the central kinases responsible for initiating tau phosphorylation is glycogen synthase kinase-3 (GSK3) and Wnt signaling play a vital role in activating GSK-3 β and GSK-3 β -mediated tau phosphorylation ¹⁴. The physiological mechanisms governing their relationship are still poorly understood.

There has been a close relationship between metabolic disorders and cognitive impairment across the AD spectrum ^{15,16}. Accounting for nearly 95% of AD patients were categorized as sporadic ones, whose increasing incidence and mortality were tightly associated with type 2 diabetes mellitus (T2DM), obesity, and hyperlipidemia ^{17,18,19}. There was about 37 percent comorbidity between AD and diabetes reported by the Alzheimer's Association Report ^{20,21}.

As a result of the strong association and share mechanism between AD and T2DM, AD was termed “Type 3 Diabetes” by some researchers^{22, 23, 24, 25}. Several studies demonstrate that diabetes confers a 1.6-fold increased risk of developing dementia^{26, 27}. Similarly, central obesity and high body mass index (BMI) during middle age have about 3.5 times increased risk of dementia later in life²⁸. Therefore, control of blood glucose and lipids is expected to be a strategy for preventing or moderating cognitive decline in aging. Nevertheless, the exact link and key associated regulator between metabolic abnormalities and AD is still unclear.

Kallistatin was a serine proteinase inhibitor, previously identified as a tissue kallikrein-binding protein²⁹. It was predominantly produced in the liver and widely expressed in the body tissues and conferred anti-angiogenesis, anti-fibrosis, anti-oxidative stress, and anti-tumor growth^{30, 31}. Furthermore, Kallistatin was found increased in patients with obesity, prediabetes, and diabetes^{32, 33, 34}. The concentration of Kallistatin was positively correlated with the triglyceride glucose index³⁵, which was proven to be an independent risk factor for dementia³⁶. In addition, our previous study revealed that the concentration of serum Kallistatin in T2DM patients was significantly increased and further clarified that Kallistatin suspended wound healing in T2DM patients by promoting local inflammation, which suggested that Kallistatin played a critical role in the progression of T2DM³⁷. Furthermore, our recent research has found that kallistatin can cause memory and cognitive dysfunction by disrupting the glutamate-glutamine cycle³⁸.

According to the relationship between T2DM, AD and Kallistatin, we constructed Kallistatin transgenic (KAL-TG) mice to explore whether it could cause cognitive impairment through up-regulation of A β production. Taken together, our results suggest that a novel regulatory mechanism of A β production and tau protein hyperphosphorylation by Kallistatin is involved in the progression of metabolic abnormalities-related AD.

Results

Kallistatin was increased in AD patients and AD model mice

To explore the relevance of AD in T2DM (**Fig. S1A**), we first performed a GAD disease enrichment analysis of differentially expressed genes in neurons of T2DM and normal controls, and found AD was closely related to T2DM (GSE161355). We further identified enrichment of the Serpin family protein domain using PFAM analysis on the David database (**Fig. S1B**) (<https://david.ncifcrf.gov/>). Our previous studies found that Kallistatin (serpin family a member 4) was elevated in the serum of T2DM patients and caused an adverse prognosis of diabetes complications⁴⁴. We have collected 11 serum samples of dementia patients from Sun Yat-sen Memorial Hospital and found that the concentration of Kallistatin was higher compared to the normal control³⁸. Here, we enrolled 56 AD patients and 61 healthy controls from four hospitals in Guangdong province to further explore the potential relevance of Kallistatin and AD. The Clinical and biochemical characteristics are presented in **Table S1**, **S2**. In addition, we measured the serum Kallistatin (12.78 ± 2.80 $\mu\text{g/ml}$) content in patients with AD was higher than in normal controls (9.78 ± 1.93 $\mu\text{g/ml}$) (**Fig. 1A**). Likewise, the fasting blood glucose (FBG) and triglyceride (TG) were more elevated in AD patients than in healthy human (**Fig. 1B**). We further grouped all the AD patients according to whether they had diabetes, then found Kallistatin (13.79 ± 3.05 $\mu\text{g/ml}$) and TG content were further elevated in AD patients with diabetes (**Fig. 1C-D**). Similarly, Kallistatin expression was increased in AD model mouse SAMP8 compared with its control mouse SAMR1 in hippocampus tissue (**Fig. S1C-D**). Taken together, these results indicated that Kallistatin concentration was increased in metabolic abnormalities-related AD patients.

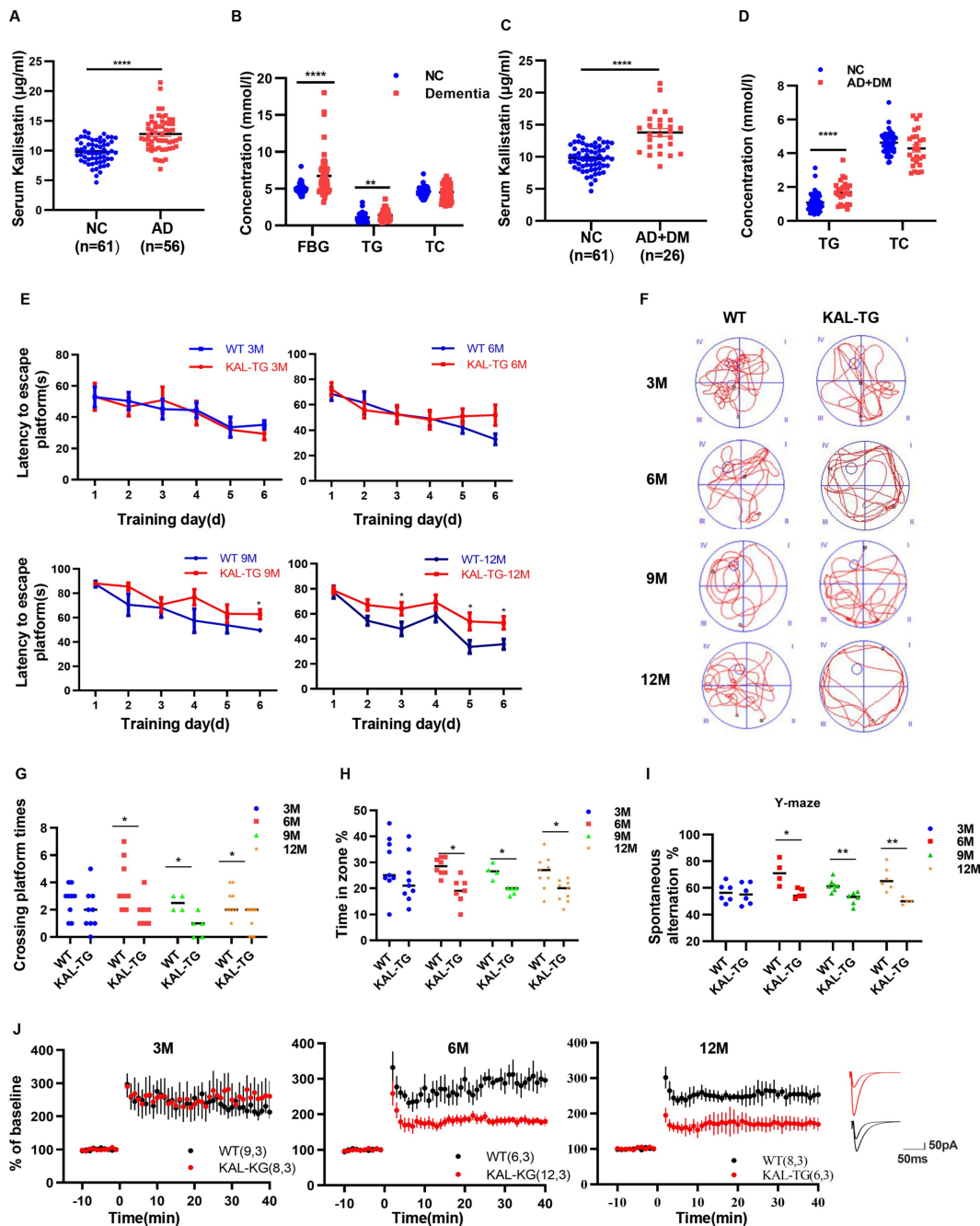


Fig. 1

Increased Kallistatin was presented in AD patients and could impair cognitive memory in mice.

(A-B) Serum Kallistatin(A), fasting blood glucose (FBG), triglyceride (TG), and total cholesterol (TC). (B) of AD patients and their corresponding normal control subjects. (C-D) Serum Kallistatin(C), TG, and TC(D) of AD patients with DM and their corresponding normal control subjects. The behavioral performance of KAL-TG mice was assessed through the Morris water maze test. (E-H), Y-maze test(I), and electrophysiology (J). (E) The escape latency time of different months of KAL-TG mice (3M, 6M, 9M, 12M) and corresponding WT mice were presented during 1-6 day. (F-H) Cognitive functions were evaluated by spatial probe test at day 7, the representative each group mice traces were shown (F), then analyzing each group mice crossing platform times (G) and time percent in the targeted area (H), $n=4$ to 9 per group. (I) Spontaneous alternation of Y-maze test. (J) LTP was measured by whole-cell voltage-clamp recordings of CA1 neurons in acute hippocampal slices of KAL-TG (3M, 6M, 12M) and WT mice, $n = 6-12$ cells from 3 mice per group. Error bars represent the standard deviation (SD); one asterisk, $p < 0.05$, two asterisks, $p < 0.01$; three asterisks, $p < 0.001$; four asterisks, $p < 0.0001$.

Kallistatin could impair cognitive memory in mice

The above experiments proved that Kallistatin increased in AD patients and AD model mice. Then, we constructed KAL-TG mice and assessed their behavioral performance through the Morris water maze (MWM) and Y-maze tests. Notably, the latency to escape platform was increased, and crossing platform times, time in percent, and spontaneous alternation were significantly decreased in KAL-TG mice compared with age-matched WT mice (**Fig. 1E-I**). Furthermore, long-term potentiation (LTP) was measured by whole-cell voltage-clamp recordings of CA1 neurons in acute hippocampal slices of KAL-TG and WT mice to determine the changes in hippocampal synapses. LTP of KAL-TG mice was significantly lower than WT mice (**Fig. 1J**). These results showed Kallistatin could impair cognitive memory in mice.

Kallistatin promoted A β deposition and tau phosphorylation

We evaluated A β deposition and tau phosphorylation in these experimental mice hippocampus tissues by immunohistochemistry staining and western blot. Predictably, the plaque density and tau phosphorylation of KAL-TG mice were much higher than age-matched WT mice (**Fig. 2A-C**, 3A-D). Consistent with this result, ELISA detection of A β 42 contents in hippocampus tissue showed that A β production was extraordinarily increased in KAL-TG mice compared to WT mice (**Fig. 2D**). These results suggested that Kallistatin promoted A β deposition and tau phosphorylation.

Kallistatin positively regulated A β generation by promoting β -secretase rather than γ -secretase

Western blot and ELISA analysis of A β levels in primary hippocampal neurons (immunofluorescence identified with neural marker MAP2, **Fig. S2G**) infected with overexpressing Kallistatin adenovirus were higher than in control groups (**Fig. 2E-G**), as well as HT22 cells (**Fig. S2A-C**). Amyloid-beta precursor protein (APP) undergoes proteolytic processing to generate peptide fragments⁴⁵, β -secretase (BACE1) and γ -secretase, composed of presenilin 1 (PS1), nicastrin, and Pen-2, were crucial enzymes for A β generation^{46, 47}. We determined the levels of APP, BACE1, and PS1 in hippocampus tissue. BACE1 protein and mRNA accumulate to higher levels in KAL-TG mice compared with WT mice (**Fig. 4A-C**, S2D), whereas no significant difference appeared in APP, PS1, and α -secretases (ADAM9, ADAM10, and ADAM17) expression (**Fig. 4A**, S2E). Consistent with the above results, the activity of BACE1 was increased (**Fig. 4D**), while PS1 activity was not changed (**Fig. S2F**). Similarly, the expression and activity of BACE1 were further determined to increase in primary mouse neurons, and HT22 cells transfected with Kallistatin adenovirus (**Fig. 5A-C**, S3A-C), and PS1 expression and activity were still not changed (**Fig. 5A**, S3D, S3A). Furthermore, the effect of Kallistatin was attenuated by the BACE1 inhibitor (verubecestat) or siBACE1 03, the most effective one (**Fig. 5E-F**, S3D). Therefore, these results indicated that Kallistatin could promote A β generation through upregulating BACE1 expression.

Kallistatin suppressed PPAR γ activation to promote BACE1 expression

Transcription factors SP1, YY1, and PPAR were reported to regulate BACE1 expression at the transcription level. Among them, PPAR γ could down-regulated BACE1 expression^{48, 49, 50}. As elaborated, PPAR γ decreased in the hippocampal tissue of KAL-TG mice detected by western blot and immunohistochemical analysis (**Fig. 5G-J**). However, no statistical differences appeared in the YY1 and SP1 expressions (**Fig. 5G**). In addition, Kallistatin downregulated the expression of PPAR γ on primary hippocampal neurons and HT22 cells, thus increasing BACE1 and A β

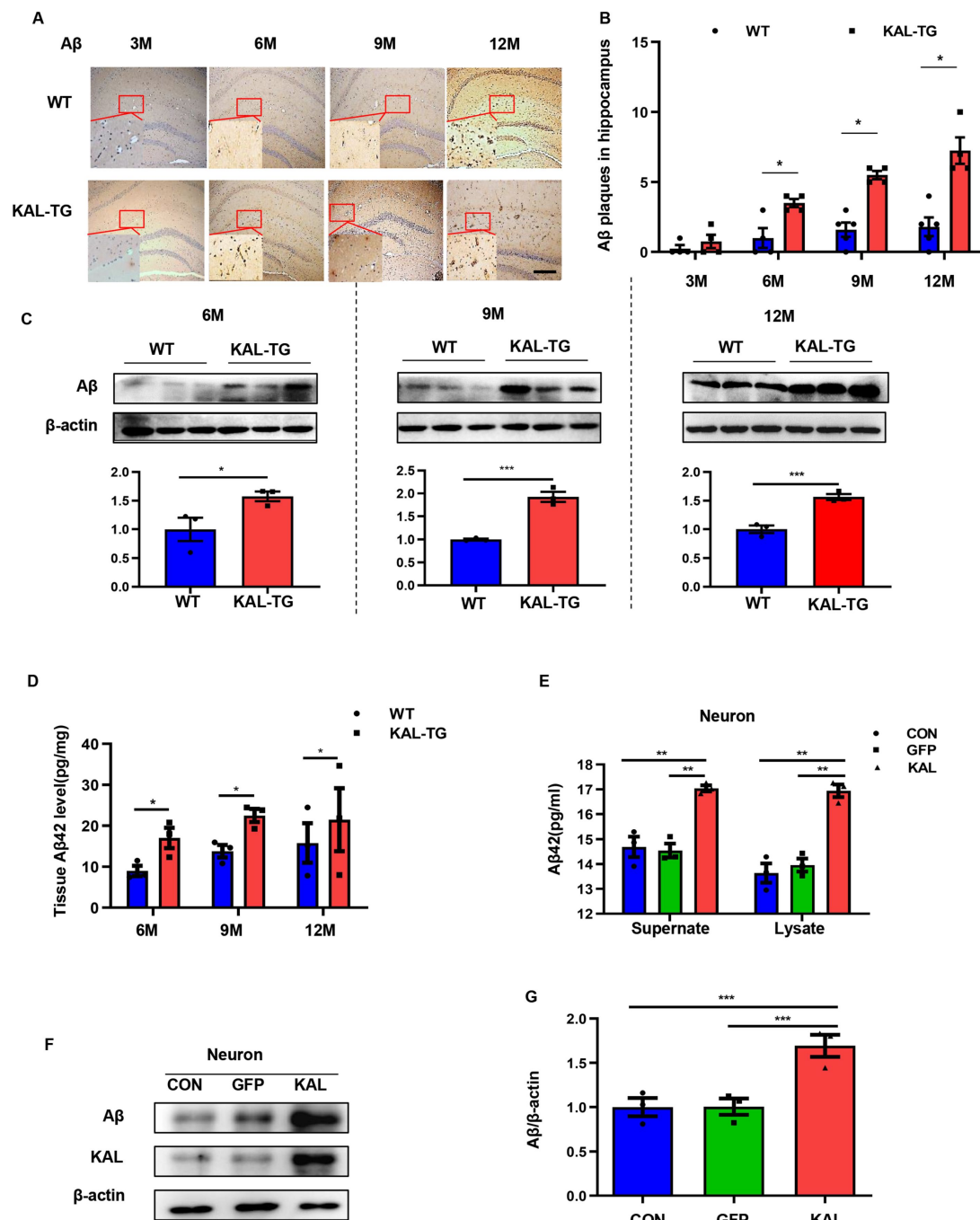


Fig. 2

Kallistatin promoted Aβ generation.

(A-B) Immunohistochemistry staining of Aβ(A) was carried out in KAL-TG and WT mice hippocampal tissue. Scale bar, 100μm. The statistical analysis of Aβ plaques (B) in hippocampal tissue of KAL-TG and WT mice, n=4-5 per group. (C) Protein levels of Aβ were tested by western blot analysis in hippocampal tissue, n=3 per group, then statistically analyzed the above results. (D) Hippocampal tissue Aβ42 contents were performed by ELISA in KAL-TG and WT groups, n=3 per group. (E) Primary mouse neurons were isolated, then infected with adenovirus to overexpress Kallistatin for 72h. Aβ42 concentration of primary hippocampal neurons supernate and cell lysate were quantified by ELISA, n=3 per group. (F-G) Western blot analysis of Aβ protein level in primary hippocampal neurons infected with overexpressing Kallistatin adenovirus and control groups, then statistical analysis of Aβ protein levels, n=3 per group. Error bars represent the standard deviation (SD); one asterisk, $p < 0.05$; two asterisks, $p < 0.01$; three asterisks, $p < 0.001$.

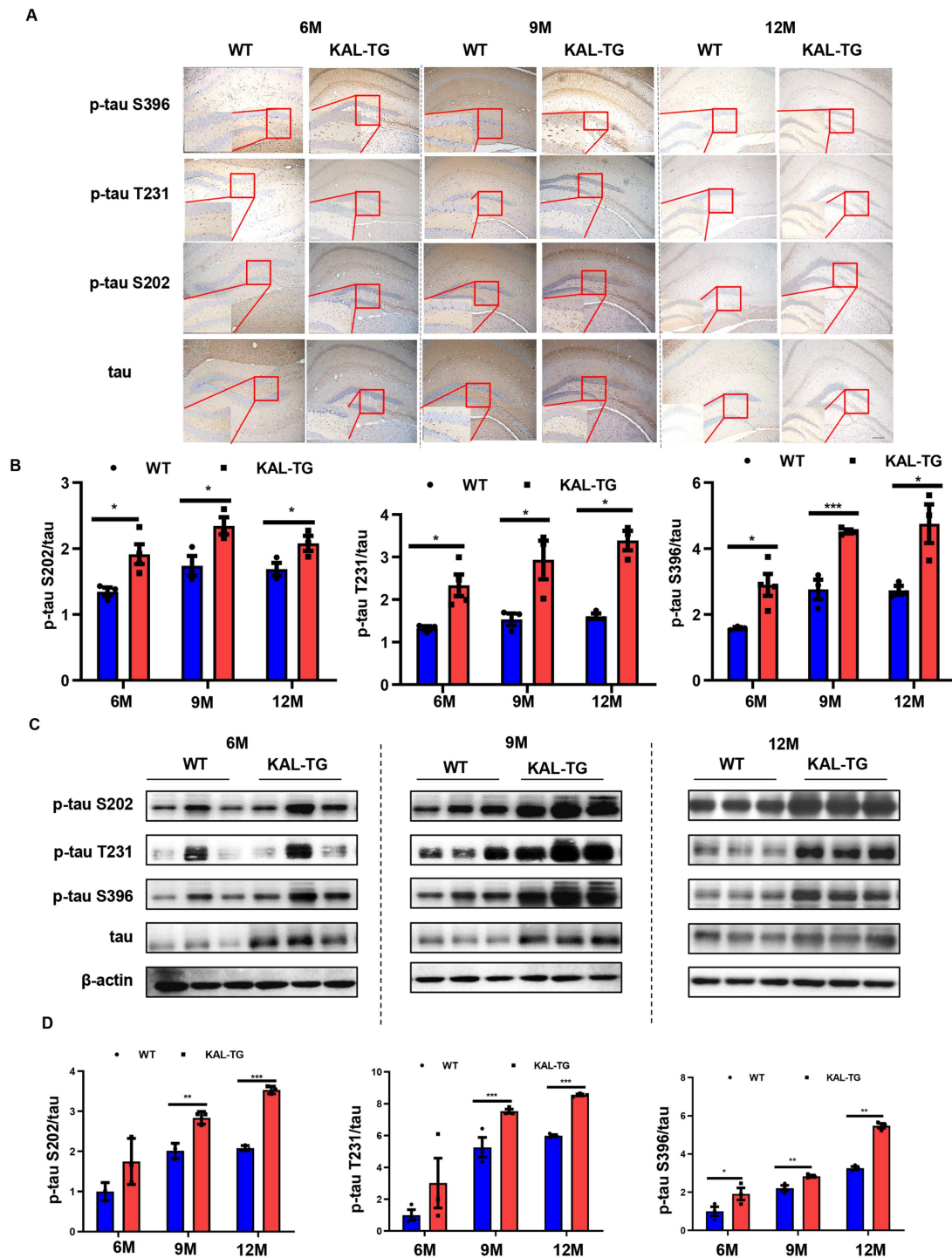


Fig. 3

Kallistatin promoted tau phosphorylation.

(A-B) Immunohistochemistry staining of phosphorylated tau (p-tau S396, p-tau T231, p-tau S202) and tau(A) was carried out in KAL-TG and WT mice hippocampal tissue. Scale bar, 100 μ m. The statistical analysis of phosphorylated tau(B) in hippocampal tissue of KAL-TG and WT mice, n=3 per group. (C-D) Protein levels of phosphorylated tau (p-tau S396, p-tau T231, p-tau S202) and tau were tested by western blot analysis in hippocampal tissue, then statistically analyzed the above results. Error bars represent the standard deviation (SD); one asterisk, $p < 0.05$; two asterisks, $p < 0.01$; three asterisks, $p < 0.001$.

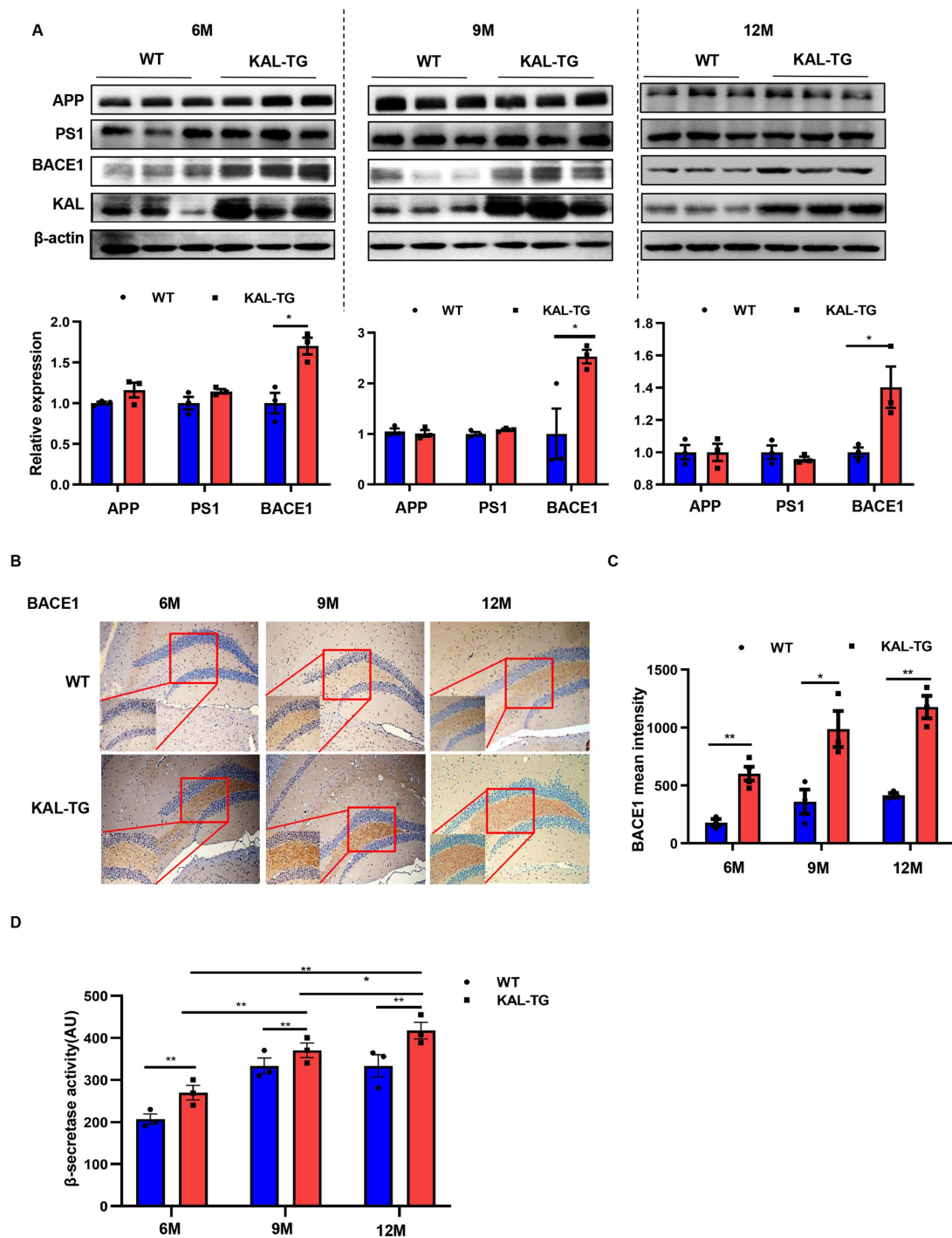


Fig. 4

Kallistatin transgenic mice exhibited increased BACE1 expression and activity in the hippocampus.

(A) Western blot analysis of relevant proteins, such as APP, PS1, and BACE1 during A β generation in hippocampal tissue of each time point (6M, 9M, 12M) KAL-TG mice and corresponding WT control groups, $n=3$ per group, then statistical analysis of APP, PS1 and BACE1 protein levels. (B) Immunohistochemistry staining of BACE1 was carried out in KAL-TG and WT mice hippocampal tissue at each time point (6M, 9M, 12M). $n=3$ to 5 per group. Scale bar, 100 μ m. (C) Statistical analysis of BACE1 immunohistochemistry staining, $n=3$ to 4 per group. (D) ELISA measured the β -secretase activity of each group's hippocampal tissue, $n=3$ per group. Error bars represent the standard deviation (SD); one asterisk, $p < 0.05$, two asterisks, $p < 0.01$.

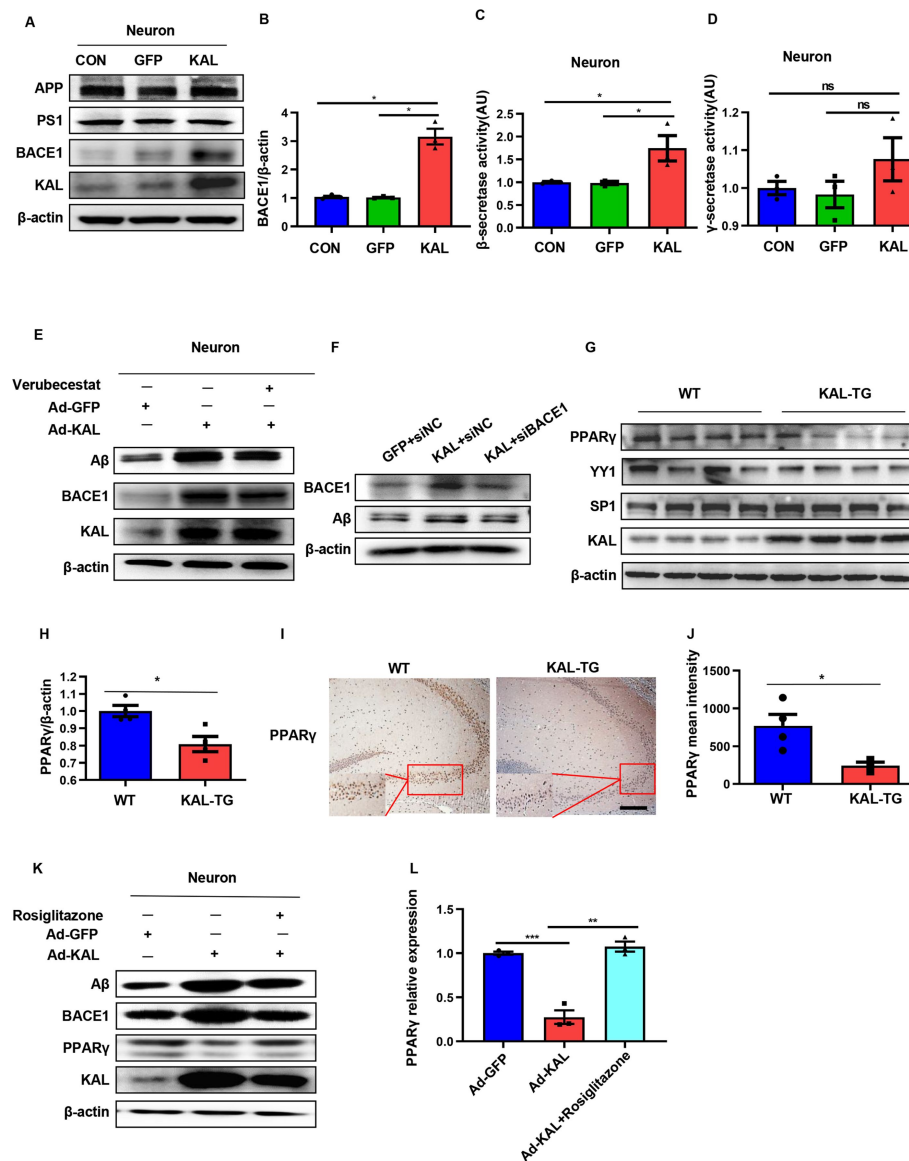


Fig. 5

***In vitro*, Kallistatin promoted BACE1 expression to augment A β by suppressing PPAR γ activation.**

(A) The relevant protein levels in primary mouse neurons infected with overexpressing Kallistatin adenovirus during A β generation were determined by western blot analysis. (B) Statistical analysis of BACE1 expression in primary neurons. (C-D) β -secretase (C) and γ -secretase (D) activity of primary hippocampal neurons infected with overexpressing Kallistatin adenovirus and control adenovirus was measured by ELISA. (E) Primary hippocampal neurons were treated with BACE1 inhibitor verubecestat (50nM), then infected with adenovirus to overexpress Kallistatin. Western blot analysis of A β , BACE1, and Kallistatin protein levels, β -actin served as a loading control. (F) HT22 cells were infected with BACE1 siRNA, then infected with adenovirus to overexpress Kallistatin. Western blot analysis of A β and BACE1 protein levels, β -actin served as a loading control. (G) The relevant proteins involved in BACE1 transcriptional expressions, such as PPAR γ , YY1, and SP1 were measured by western blot analysis in hippocampal tissue. β -actin served as a loading control. (H) Statistical analysis of PPAR γ in hippocampal tissue of each group. (I) The representative diagrams of PPAR γ expression in hippocampal tissue were presented in the above graphs. Scale bar, 100 μ m. (J) Statistical analysis of PPAR γ immunohistochemistry staining in hippocampal tissue of each group, n=3 to 4 per group. (K) Primary hippocampal neurons were treated with PPAR γ agonist rosiglitazone (10nM) for 12h, then infected with adenovirus to overexpress Kallistatin for 72h. Western blot analysis of A β and BACE1 protein levels. β -actin served as a loading control. (L) Statistical analysis of PPAR γ protein levels in each group. Error bars represent the standard deviation (SD), one asterisk, $p < 0.05$, two asterisks, $p < 0.01$; three asterisks, $p < 0.001$; ns means no significant difference.

expression (Fig. 5K [↗](#), S3E). Treatment with rosiglitazone, the PPAR γ agonist, could reverse the decrease of PPAR γ caused by Kallistatin (Fig. 5K [↗](#)-L). Predictably, rosiglitazone could inhibit the promoting effect of Kallistatin on BACE1 and A β (Fig. 5K [↗](#) and Fig. S3E [↗](#)).

Kallistatin promoted A β production *via* directly binding to the Notch1 receptor and activating the Notch1 pathway

Our results indicated that Notch1 was highly expressed in the hippocampal tissues of KAL-TG mice (Fig. 6A [↗](#)-D). Moreover, primary hippocampal neurons and HT22 cells infected with overexpressing Kallistatin adenovirus had upregulated Notch1 expression *in vitro* (Fig. S4A-B [↗](#)). In addition, Kallistatin could directly bind to the Notch1 receptor and activate the Notch1 pathway (Fig. 6E [↗](#)-F and Fig. S4C-D [↗](#)). Treatment with the siNotch1 03, which was the most effective (Fig. S4E [↗](#)), to knock down Notch1 inhibited the effect of Kallistatin on the activation of the Notch1 signaling pathway, resulting in the downregulation of HES1, upregulation of PPAR γ , and downregulation of BACE1 and A β (Fig. 6G [↗](#)). HES1, an essential downstream effector of Notch1 signaling pathway, was reported to suppress the expression of *PPARG* (gene name of PPAR γ) in neuron cells [51](#)[↗](#), [52](#)[↗](#). Similarly, the HES1 shRNA 01, the most effective (Fig. S4F [↗](#)), could upregulate PPAR γ and decrease the production of BACE1 and A β when the neuron cells are infected with adenovirus to overexpress Kallistatin (Fig. 6H [↗](#)). The above results suggested that Kallistatin promotes A β production *via* directly binding to the Notch1 receptor and activating the Notch1 pathway.

Kallistatin promoted the phosphorylation of tau by activating the Wnt signaling pathway

Glycogen synthase kinase 3- β (GSK-3 β) is a crucial element in the phosphorylation of tau [53](#)[↗](#). When the Wnt signaling was activated, LRP6 PPPSPxS motif could directly interact with GSK3- β and phosphorylate it [54](#)[↗](#). So when the Wnt signaling was inhibited, GSK3 β was activated and dephosphorylated, and non-phosphorylated GSK3 β was able to add phosphate groups to serine/threonine residues of tau [14](#)[↗](#). Kallistatin has already been reported as a competitive inhibitor of the canonical Wnt signaling pathway [55](#)[↗](#). Consistent with previous reports, our results showed that GSK-3 β was activated in the hippocampus of KAL-TG mice (Fig. 7A [↗](#)-B). Furthermore, visible increase in tau phosphorylation with the activation of GSK-3 β stimulated by Kallistatin overexpression (Fig. 7C [↗](#)-D), which could be reversed by LiCl, the inhibitor of GSK-3 β (Fig. 7E-F [↗](#)). These results proved that Kallistatin could promote the phosphorylation of tau by activating the Wnt signaling pathway.

Fenofibrate could alleviate memory and cognitive impairment of KAL-TG mice

Hyperlipidemia and hyperlipidemia were account for the development of AD [56](#)[↗](#), [57](#)[↗](#). Here, a hypolipidemic drug (fenofibrate) and a hypoglycemic drug (rosiglitazone) were used to treat the KAL-TG mice (Fig. 8A [↗](#)). After a-month treatment with fenofibrate, the behavioral performance measured by MWM and Y-maze test of treated group was improved compared to KAL-TG group. The latency to the escape platform on the fifth training day was significantly decreased (Fig. 8B [↗](#)), and crossing platform times (Fig. 8C [↗](#)), time in percent (Fig. 8D [↗](#)), and spontaneous alternation (Fig. 8F [↗](#)) were significantly increased in the fenofibrate-treated group, which performed better than rosiglitazone-treated group. Similarly, the path traces heatmap indicated mice stayed in the target quadrant longer in the fenofibrate-treated group (Figure.8E). Besides, the decreased serum Kallistatin level, A β , BACE1, phosphorylation of tau, and the activation of GSK3 β was detected in the fenofibrate-treated KAL-TG group (Fig. 8G [↗](#)-I). However, no significant difference showed in the rosiglitazone-treated group compared with the KAL-TG mice (Fig. 8G [↗](#)-I).

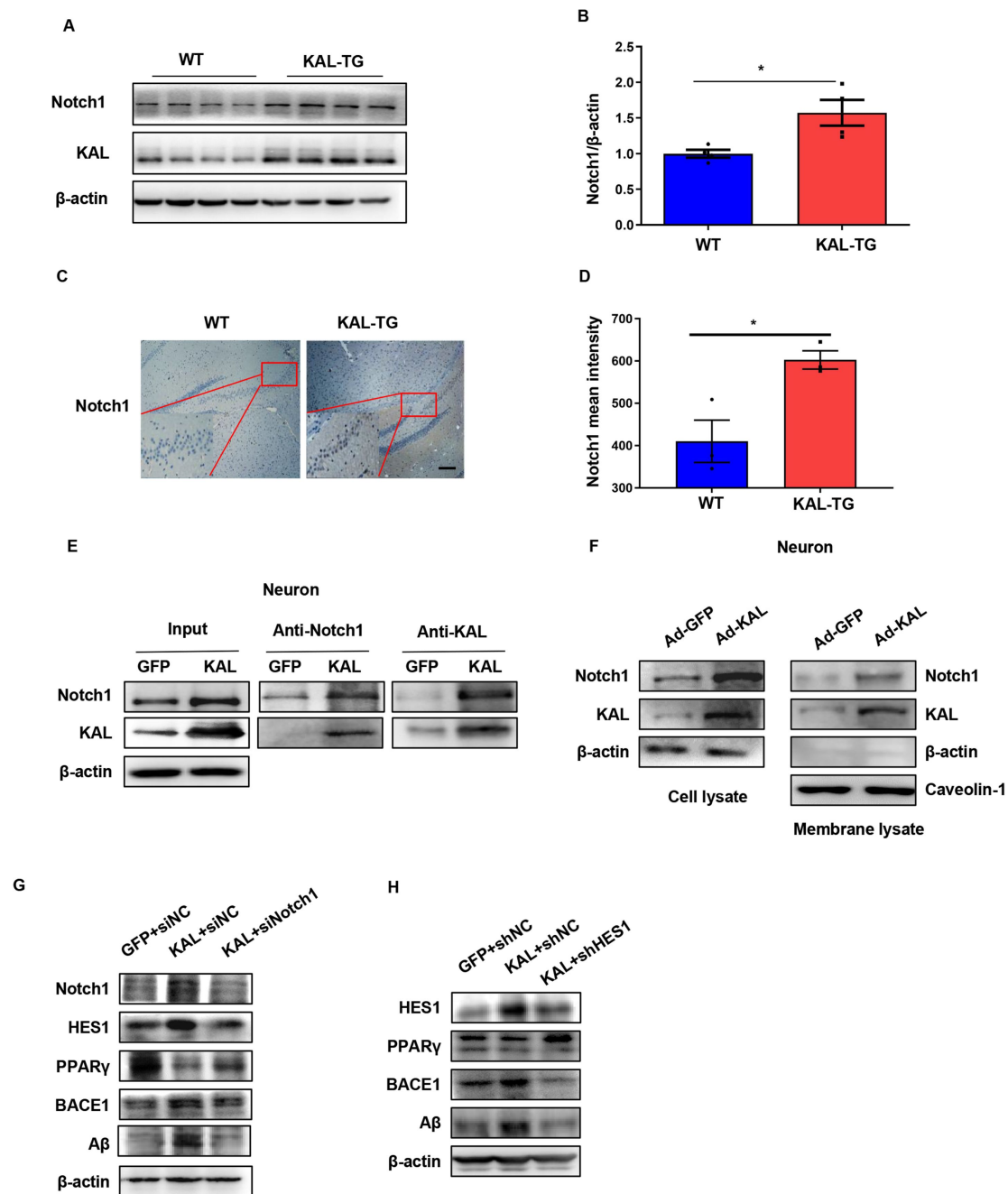


Fig. 6

Kallistatin directly bonded to the Notch1 receptor, which activated the Notch1 pathway to promote Aβ production.

(A) Notch1 expression was measured by western blot analysis in hippocampal tissue. β-actin served as a loading control. (B) Statistical analysis of Notch1 in hippocampal tissue of each group. (C) The representative diagrams of Notch1 expression in hippocampal tissue were presented in the above graphs. Scale bar, 100μm. (D) Statistical analysis of Notch1 immunohistochemistry staining in hippocampal tissue of each group. (E-F) Primary hippocampal neurons were infected with overexpressing Kallistatin adenovirus for 72h, then Co-IP analysis (E) and membrane extraction experiment (F) was performed to verify whether Kallistatin can bind to the Notch1 receptor. β-actin served as a loading control. (G-H) HT22 cells were treated with siRNA (Notch1) and shRNA (HES1) to knock down Notch1 and HES1 for 12h, then infected with adenovirus to overexpress Kallistatin for 24h. Western blot analysis was used to detect the Notch1 signaling pathway. Error bars represent the standard deviation (SD), one asterisk, $p < 0.05$.

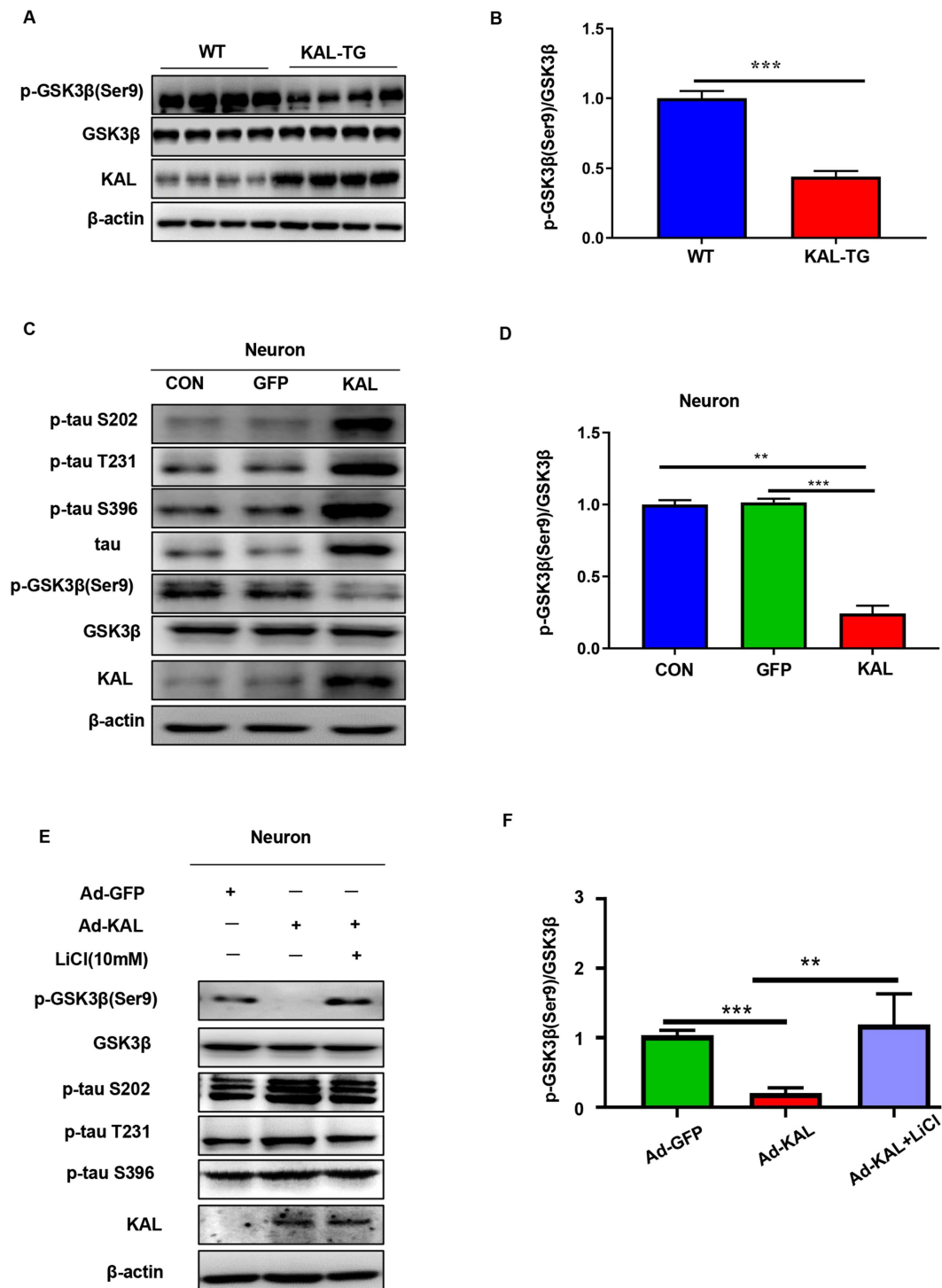


Fig. 7

Kallistatin promoted phosphorylation of tau by suppressing Wnt signaling pathway.

(A-B) GSK-3β and p-GSK-3β expression was measured by western blot analysis in hippocampal tissue, then statistically analyzed the above results. (C-D) Primary hippocampal neurons were treated with overexpressing Kallistatin adenovirus for 72h, then western blot analysis was used to detect the content of GSK-3β, p-GSK-3β, tau, p-tau (Ser9, T231, S396), and statistically analyzed the above results. (E-F) Primary hippocampal neurons were treated with overexpressing Kallistatin adenovirus for 48h, then treated with LiCl (10mM) for 24h, western blot analysis was used to detect the content of GSK-3β, p-GSK-3β, p-tau (Ser9, T231, S396), and statistical analysis of the above results.

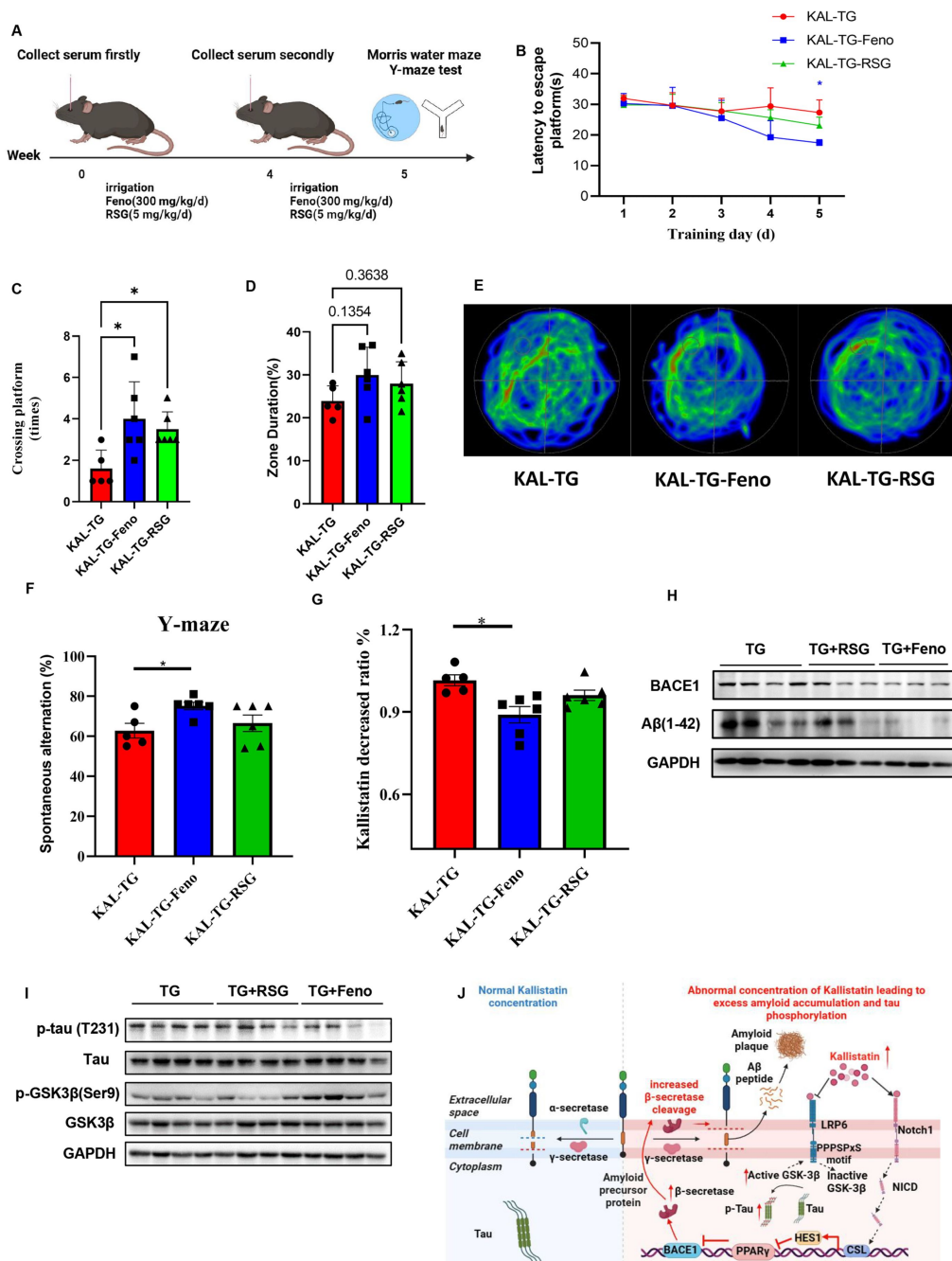


Fig. 8

Fenofibrate could alleviate memory and cognitive impairment of KAL-TG mice.

(A) Illustration of experimental protocols. Fenofibrate (0.3 g/kg/d × 5week, i.g.) or rosiglitazone (5mg/kg/d × 5week, i.g.) were given to KAL-TG mice. The serum for Kallistatin measuring was collected at week 0 and week 4. And Morris water maze and Y-maze test were performed at week 5. (B-E) Behavioral performance was assessed through the Morris water maze test (B-E) and the Y-maze test (F). (B) The escape latency time was presented during 1-5 day. (C-E) Cognitive functions were evaluated by spatial probe test at day 6, then analyzing each group of mice crossing platform times(C), time percent in the targeted area (D), and the path traces heatmap (E), n=5 to 6 per group. (F) Spontaneous alternation of Y-maze test. (G) Kallistatin decreased ratio was calculated by dividing the serum Kallistatin concentration of KAL-TG mice before Fenofibrate/ rosiglitazone treatment by the serum Kallistatin concentration of KAL-TG mice after a-month treatment, and serum Kallistatin concentration was measured by ELISA. (H) Protein levels of Aβ, BACE1, p-tau, tau, p-GSK-3β and GSK-3β were tested by western blot analysis in hippocampal tissue. Error bars represent the standard deviation (SD); one asterisk, $p < 0.05$.

Mechanism summary

In metabolic abnormalities-related AD patients, the concentration of Kallistatin is elevated, which could increase A β deposition through Notch1/HES1/PPAR γ /BACE1 pathway, and induce tau hyperphosphorylation by activating GSK-3 β . Finally, elevated Kallistatin caused impaired cognitive memory via inducing A β deposition and tau hyperphosphorylation (**Fig. 8J** [↗](#)).

Discussion

This study demonstrated that Kallistatin is a novel regulator of amyloid- β plaque accumulation, tau protein hyperphosphorylation, and metabolic abnormalities-related cognitive memory impairment. We proved that Kallistatin was increased in the serum of AD and diabetes-related AD patients and the hippocampus of AD model mice. In addition, the constructed KAL-TG mice defined its cognitive memory impairment phenotype and lower LTP in hippocampal CA1 neurons accompanied by increased A β deposition and tau phosphorylation. Mechanistically, Kallistatin transcriptionally upregulates BACE1 expression by suppressing the transcriptional repressor PPAR γ , resulting in A β cleavage and production. Most importantly, our studies proved Kallistatin could bind to the Notch1 receptor directly and activate the Notch1/HES1 pathway, causing the decrease of PPAR γ , overproduction of BACE1, and elevated A β 42 generation. Moreover, Kallistatin could induce tau phosphorylation by activating GSK-3 β , which result from the inhibition of LRP6. Finally, prolonged stimulation with high concentrations of Kallistatin could impair cognitive memory in mice. Lastly, the hypolipidemic drug-fenofibrate decreasing of A β expression, phosphorylation of tau, and the serum Kallistatin level of KAL-TG mice, causing alleviating memory and cognitive impairment. For the first time, these observations established an association between high Kallistatin levels and metabolic abnormalities-related AD, and provided a new drug candidate (fenofibrate) for AD patients with metabolic syndromes.

Growing shreds of evidence suggest that Diabetes mellitus and AD are closely linked. Approximately 80% of AD subjects are insulin resistant or T2DM [58](#)[↗](#); additionally, the T2DM patients had a higher risk of up to 73% of dementia than the healthy control population [26](#)[↗](#). In line with these observations, the process of cognitive decline in T2DM patients appears to begin in the prediabetic phase of insulin resistance [59](#)[↗](#), [60](#)[↗](#). We performed a GAD disease enrichment analysis of differentially expressed genes in neurons of T2DM and normal controls, which showed AD is closely related to T2DM, besides identified enrichment of the Serpin family protein domain using PFAM analysis (**Fig. S1A, B** [↗](#)). We and other researchers found that Kallistatin (which belongs to the Serpin family) was increased in T2DM [32](#)[↗](#), [37](#)[↗](#). Although the interrelationship between Kallistatin and AD has not been reported to date, we have enough reason to speculate Kallistatin might be one of the critical molecules which could establish the relationship between Alzheimer's disease and Diabetes. Our data suggested that AD patients indeed have metabolic disorders and elevated Kallistatin (**Fig. 1A** [↗](#)-D, S1C-D). In addition, memory cognitive function and synaptic plasticity of KAL-TG mice were impaired (**Fig. 1E** [↗](#)-J). These results suggested that Kallistatin was a new link between AD and diabetes.

A hallmark of AD is the aggregation of A β into amyloid plaques and tau phosphorylation in patients' brains. A β , a small peptide with a high propensity to form aggregates, is widely believed to be central and initial to the pathogenesis of the disease [61](#)[↗](#). Correspondingly, we discovered Kallistatin could lead to A β overproduction through Notch1/HES1/PPAR γ /BACE1 signaling pathway. GSK-3 β , a vital kinase regulating the process of tau phosphorylation [53](#)[↗](#), could be activated by inhibiting the Wnt signaling pathway. Kallistatin was already promoted as a competitive inhibitor

of LRP6, the Wnt receptor ⁵⁵. Consistent with previous reports, Kallistatin could increase tau phosphorylation by activating GSK-3 β . We demonstrated for the first time that Kallistatin could promote AD by increasing A β production and tau phosphorylation in the central nervous system.

Previous studies have shown that Notch signaling is closely related to AD. For example, NOTCH mutation was reported to cause AD-like pathology ⁶². And Notch1 was found to increase in AD patients ⁶³. Besides, the Notch1/HES1 signaling pathway was reported to suppress the expression of PPAR γ ⁶⁴. Our previous studies proved that Kallistatin could activate Notch1 signaling ³⁷. Consequently, we detect Notch1 in our animal and cell models. Indeed, Notch1 was upregulated by Kallistatin (**Fig. 6**, S4), as well as A β deposition. Notch signaling is initiated by receptor-ligand interaction at the cell surface. In mammals, there are five ligands encoded by JAG1, JAG2, DLL1, DLL3, and DLL4 ⁶⁵. Here, our results found that Kallistatin could activate Notch1 by binding to it directly, which means Kallistatin was a new ligand of the Notch1 receptor.

Treatment of AD has always been a hot and challenging problem in neurology. Multiple strategies have been proposed to reduce the pathogenicity of A β and tau. Unfortunately, several A β -targeted therapies tested in the phase III clinical trials have failed as they have not been able to slow down cognitive decline, although they could effectively reduce the A β load ^{66, 67, 68, 69, 70, 71}. BACE1 inhibitors were failed to improve the cognitive function of AD patients but also resulted in clinical deterioration and impairment of liver function ^{72, 73, 74}. Here are two possible reasons for the failure of clinical trials with the BACE1 inhibitor. First, the reduction of BACE1 activity could lead to an accumulation of full-length APP ⁷⁵. Second, the size of the BACE1 active site is relatively large; the use of a small molecule may not be sufficient to occupy the active site ⁷⁶. Therefore, synaptic damage caused by BACE1 inhibitors or the insufficient effect of BACE1 inhibitors may lead to the failure of clinical trials. Therapeutic strategies targeting tau include tau aggregation blockers (TRx0014, TRx0237), antibody vaccine therapy (e.g., RO7105705, BIIB092), inhibition of tau phosphorylation (Anavex2-73), and microtubule stabilizers (Anavex2-73) ⁷⁷. Some of the aforementioned drugs have been partially stopped, while others are still undergoing a clinical test and shown protective benefits. There are still several obstacles to the commercialization of tau treatments when they reach maturity.

Because of the failures of clinical trials, some researchers propose an alternative option for AD therapeutics to address the modifiable risk factors for developing AD, such as type 2 diabetes ^{78, 79, 80}. Our previous study proved that Kallistatin is a multifunctional protein strongly associated with diabetes, and Kallistatin neutralizing antibody improves diabetic wound healing ⁴⁴. This study demonstrated that Kallistatin induced memory cognitive dysfunction by promoting A β deposition and tau phosphorylation. Thus, we speculate the increased Kallistatin is a decent candidate for T2DM-related AD therapy. PPAR γ is a ligand-activated transcription factor and a master modulator of glucose and lipid metabolism, organelle differentiation, and inflammation ^{81, 82}. Growing evidence revealed that PPAR γ agonists (rosiglitazone) could rescue memory impairment of AD model mice ^{83, 84, 85}. In clinical trials, it is controversial whether rosiglitazone has a protective effect on memory cognitive function ^{86, 87, 88}. In this study, although A β expression had a downward trend, the memory and cognition of KAL-TG mice were unchanged after treatment with rosiglitazone for a month (**Fig. 8**). This might be caused by insufficient treatment time and the unchanged Kallistatin level.

Fenofibrate is a fibric acid derivative for clinically lowering blood lipids, mainly triglycerides ⁸⁹. Studies showed that fenofibrate could prevent memory disturbances, maintain hippocampal neurogenesis, and protect against Parkinson's disease (PD) ^{90, 91}. Specifically, fenofibrate has a neuroprotective effect on memory impairment induced by A β through targeting α - and β -secretase ⁹². Recently, our study proved that the fenofibrate could repair the disrupted glutamine-glutamate cycle by upregulating glutamine synthetase, while there is currently no fenofibrate treatment of AD in clinical trials ³⁸. In this study, we proved fenofibrate was beneficial for memory and cognitive impairment of KAL-TG mice. In addition, A β , BACE1, phosphorylated tau

and serum Kallistatin level of KAL-TG mice could be downregulated after the treatment of fenofibrate. All of these suggested that fenofibrate might be helpful for metabolic abnormalities-related AD patients. Therefore, fenofibrate administration in patients with metabolic syndrome play an early role in preventing and treating AD.

In summary, we affirmed that Kallistatin concentrations were increased in diabetes-related AD patients. In addition, our study demonstrated for the first time that Kallistatin positively regulated A β 42 through Notch1/HES1/PPAR γ /BACE1, and increased phosphorylated tau through inhibition of the Wnt signaling pathway. Kallistatin might play a crucial role in linking diabetes and cognitive memory deterioration. Moreover, fenofibrate could decrease the serum Kallistatin level, BACE1, A β , and phosphorylated tau of KAL-TG mice, leading to alleviating memory and cognitive impairment. These findings might provide new insight into AD and possibly other neurodegenerative disorders.

Materials and methods

Ethics approval and consent to participate

All patients involved in this study gave their informed consent. This study obtained the institutional review board approval of Medical Ethics of Zhongshan Medical College No. 072 in 2021 and Animal Experiment Ethics of Zhongshan Medical College, Approval No.: SYSU-IACUC-2019-B051.

Human Subjects

The study was approved by the experimental ethics committee of Guangdong Academy of Medical Sciences and Sun Yat-sen University and carried out in strict accordance with the ethical principles, and each participant was provided written informed consent before collecting samples. We certify that the study was performed in accordance with the 1964 declaration of HELSINKI and later amendments. We collected 61 normal human samples, 56 AD patient samples, of whom 36 normal human samples were from the Zhongshan City People's Hospital; 14 normal human samples and 22 AD patient samples were from Zhongshan Third People's Hospital; 11 normal human samples and 14 AD patient samples were from Sun Yat-sen Memorial Hospital. AD patient was clinically diagnosed according to ICD-10 (International Classification of Diseases) and NINCDS-ADRDA (the National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association) criteria, and 20 AD patient samples were clinically diagnosed according to MMSE (Mini-Mental State Examination), were collected from Guangdong Provincial People's Hospital. All subjects' Clinical characteristics were presented in Supplementary Tables (**Table S1** [↗](#)-**2** [↗](#)).

Experimental Animals and Protocols

All animal experiment procedures were carried out in an environment without specific pathogens (Specific pathogen free, SPF) with the approval of the Animal Care and Use Committee of Sun Yat-sen University (approval ID: SYXK 2015-0107). The wild type mice (WT, C57BL/6) were purchased from the Animal Center of Guangdong Province (Production license No.: SCXK 2013-0002, Guangzhou, China). The SAMR1 and SAMP8 mice (7 months old) were purchased from Tianjin University of Traditional Chinese Medicine (Tianjin, China). Kallistatin transgenic mice (KAL-TG) were C57BL/6 strain provided by Dr. Jianxing Ma (University of Oklahoma Health Sciences Center) [39](#) [↗](#). The KAL-TG mice genotype was identified by PCR technology (forward primer: 5'-AGGGAAG-ATTGTGGATTGG-3', reverse primer: 5'-ATGAAGATACCAGTGATGCTC-3'). KAL-TG mice aged 6 months were randomly divided into three groups: control group (KAL-TG), fenofibrate-treated group (KAL-TG-Feno, 0.3 g/kg/d), and rosiglitazone-treated group (KAL-TG-RSG, 0.005 g/kg/d). Fenofibrate (Sigma-Aldrich, cat. no. F6020) and rosiglitazone (Selleck, cat. no. S2556) were

administered to mice by oral gavage. In three groups, the serum Kallistatin were examined in the 0 week and 4 week after drug treatment from the blood taken from mouse orbit. In addition, the Morris water maze and Y-maze test were performed one week after the second blood collection.

Morris water maze (MWM)

The KAL-TG and WT mice were employed for the Morris water maze test including the behavioral test, latency experiment (for 6 days), and the probe test (the 7th day). In addition, the MWM was performed as described previously⁴⁰. Mice were brought into the testing room and handled for 1 day before the training experiment. In the 6-day training experiment, each mouse was trained with four daily trials. The mice facing the wall were placed into the maze, exploring the maze from different directions (east, south, west, and north). This trial was completed as soon as the mouse found the platform, or 90 s elapsed. If the mice could discover and climb the submerged platform within 90 s, the system would automatically record the latency time and path immediately, and then the mouse was guided to and placed on the submerged platform for extra 20 s. On day 7, the platform was removed, and a probe test was performed to examine the strength and integrity of the animal spatial memory 24 h after the last testing trial. During the probe test, the mice were gently brought into the water from the fixed monitoring point, and the mice were allowed to swim for 90 s without the platform. Finally, all of the measured behavioral parameters were analyzed using SMART software.

Y-maze test

A Y maze test was performed to assess the mice's spatial memory. The Y maze was separated by 120°, consisting of three identical arms (30 cm long, 7 cm wide, and 15 cm high) made of blue PVC. The mice were placed first in one of the arms, and over the next 10 minutes, the sequence and number of their entry into the three arms were monitored. An alternation is defined when a mouse visits three straight arms (namely, ABC, BCA, or CAB, but not ABA, BAB, or CAC). Spontaneous alternation (%) = [(number of alternations)/(total number of arms-2)] × 100.

Electrophysiology

Hippocampal slices (300-400 μm) from KAL-TG and WT mice were cut as described⁴¹. Coronal slices from hippocampus (400 μm thick) were prepared from different age groups KAL-TG mice and their WT littermates using a tissue slicer (Vibratome 3000; Vibratome) in ice-cold dissection buffer containing the following (in mM): 212.7 sucrose, 3 KCl, 1.25 NaH₂PO₄, 3 MgCl₂, 1 CaCl₂, 26 NaHCO₃, and 10 dextrose, bubbled with 95% O₂/5% CO₂. The slices were immediately transferred to ACSF at 35 °C for 30 min before recordings. The recipe of ACSF was similar to the dissection buffer, except that sucrose was replaced with 124 mM NaCl, and the concentrations of MgCl₂ and CaCl₂ were changed to 1 mM and 2 mM, respectively. All recordings were performed at 28-30 °C. Pyramidal cells in CA1 areas were identified visually under infrared differential interference contrast optics based on their pyramidal somata and prominent apical dendrites. Whole-cell was performed using an Integrated Patch-Clamp Amplifier (Sutter Instrument, Novato, CA, USA) controlled by Igor 7 software (WaveMetrics, Portland, OR, USA) filtered at 5 kHz and sampled at 20 kHz. Igor 7 software was also used for acquisition and analysis. Only cells with series resistance <20 MΩ and input resistance >100 MΩ were studied. Cells were excluded if input resistance changed >15% or series resistance changed >10% over the experiment. A concentric bipolar stimulating electrode with a tip diameter of 125 μm (FHC) was placed in the stratum radiatum. The recording and stimulating electrode distances were kept at 50-100 μm. Patch pipettes (2-4 MΩ) were filled with the internal solution consisting of the following (in mM): 120 Cs-methylsulfonate, 10 Na-phosphocreatine, 10 HEPES, 4 ATP, 5 lidocaine N-ethyl bromide (QX-314), 0.5 GTP; the pH of the solution was 7.2-7.3, and the osmolality was 270-285 mOsm.

To induce LTP, a pairing protocol was applied. In brief, conditioning stimulation consisted of 360 pulses at 2 Hz paired with continuous postsynaptic depolarization (180 s) to 0 mV. 50 μ M picrotoxin was added to the recording bath to suppress excessive polysynaptic activity, and the concentration of Ca^{2+} and Mg^{2+} was elevated to 4 mM to reduce the recruitment of polysynaptic responses. A test pulse was delivered at 0.067 Hz to monitor baseline amplitude for 10 min before and 30 min following paired stimulation. To calculate LTP, the EPSC amplitude was normalized to the mean baseline amplitude during 10 min baseline. Potentiation was defined as the mean normalized EPSC amplitude 25–40 min after paired stimulation.

ELISA

To quantify serum Kallistatin, the collected samples were centrifuged at 4 °C for 10 min at 5000 rpm. It was detected using the KBP ELISA kit (#DY1669, R&D systems, MN, USA) as per the instructions of the manufacturer. The levels of A β 42 in brain tissue produced from mouse primary neuron cells and HT22 cells were measured with a mouse A β 42 Elisa Kit (27721, IBL, Germany). To measure A β 42 in brain tissue, 0.05 g of mouse brain tissues were weighed and homogenized using 2 ml PBS with a protease inhibitor (cocktail). After centrifugation at 4 °C for 30 min at 12000 g, the extracts' supernatants were analyzed using the ELISA method after total protein quantification. To quantify levels of A β 42 produced from primary neuron cells, the cell supernatants were ultrafiltrated with an Ultrafiltration tube (4-kD Millipore), centrifugation, and testing. Cell homogenate was prepared in 1 ml PBS with a protease inhibitor (cocktail) and quantified using the BCA method before being measured by ELISA.

Immunohistochemistry

Tissue slices were prepared as described before ⁴². The sections were incubated with A β (ab201060, Abcam, Cambridge, UK), BACE1 (#5606S, Cell Signaling Technology, Boston, USA), PPAR γ (#2435, Cell Signaling Technology, Boston, USA), Notch1 (#3608, Cell Signaling Technology, Boston, USA), p-tau S202 (ab108387, Abcam, Cambridge, UK), p-tau T231 (ab151559, Abcam, Cambridge, UK), p-tau S396 (ab109390, Abcam, Cambridge, UK), tau (ab75714, Abcam, Cambridge, UK) antibodies overnight at 4 °C and then incubated with Alexa Fluor 488-donkey anti-rabbit IgG (H + L) (1:200, Life Technologies, Gaithersburg, MD, USA, #A21208) for 1 h, then incubated with a biotin-conjugated secondary antibody for 30 min, followed by incubation with DAB for 10 s and hematoxylin staining for 30 s. The IHC signals were analyzed using ImageJ.

Cell culture experiments

HT22 cells were purchased from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China). HT22 cells were cultured and grown to confluence in DMEM supplemented with 10% FBS (Gibco BRL), 100 U/mL penicillin, and 100 U/mL streptomycins (Gibco BRL).

Primary culture of hippocampal neurons

Primary neurons were obtained from the hippocampus of C57/BL6J mice at age 1–3 days. Before culturing, the newborn pup was euthanized and dipped into 70% ethanol for 3 min. First, the infant pup hippocampus was isolated with eye tweezers observed under the stereomicroscope, and excess soft tissue was removed. Second, hippocampal tissue in PBS buffer was cut up with scissors gently and blown with a 1 ml pipette until it was not visible. Next, the cell suspension was transferred to a 15 ml centrifuge tube and centrifuged at 1000 rpm for 5 min at room temperature. Cell precipitation was suspended and cultured with 2–3 mL primary neural stem cell (NSC) suspension (Thermo Fisher Scientific, 21103049) in a 37 °C, 5% CO₂ cell incubator for 3 days, changing half medium every 2 days. After 7 days, the cell suspension was transferred to a 15 ml centrifuge tube, centrifuged, and recultured with neurobasal, 10% FBS, 1:50 B27 (Thermo Fisher Scientific, A3582801), and 1:100 bFGF (Thermo Fisher Scientific, #RP-8626). One day later, the

medium was changed to neurobasal (2%FBS, 1:50 B27, and 1:100 bFGF), culturing for 21 more days. The immunofluorescence technique was used with the neuron-specific marker (MAP2, #4542, Cell Signaling Technology, Boston, USA) to determine the purity of neurons.

siRNA, shRNA, and adenovirus transfection

Notch1 siRNA and control siRNA were purchased from RiboBio (Guangzhou, China). HES1 shRNA and control shRNA were purchased from Qingke (Guangzhou, China). Green fluorescent protein-adenovirus (Ad-GFP) and Kallistatin-adenovirus (Ad-KAL) were provided by Dr. Jianxing Ma (University of Oklahoma Health Sciences Center). According to the manufacturer's instructions, the transfections were performed at approximately 60% confluency using Lipofectamine®3000 transfection reagent (Invitrogen) or RNAiMAX. After 24 h, interference confirmation was conducted using real-time quantitative PCR (RT-qPCR) and Western blot.

RNA isolation and quantitative RT-PCR

Total RNA extraction, reverse transcription of cDNA, and real-time quantitative PCR were performed as described previously⁴³. BACE1 forward: GGAGCCCTTCTTTGACTCCC; BACE1 reverse: CAATGATCATGCTCCCTCCA; ADAM9 forward: GGAAGGCTCCCTACTCTCTGA; ADAM9 reverse: CAATTCC-AAACTGGCATTCTCC; ADAM10 forward: ATGGTGTTGCCGACAGTGTTA; ADAM10 reverse: GTTTGGCAGCTGGTGTCTTTT; ADAM17 forward: GGAT-CTACAGTCTGCGACACA; ADAM17 reverse: TGAAAAGCGTTTCGGTACTTGAT; β -actin forward: GCACTCTCCAGCTTCCTT; β -actin reverse: GTTGCGTACAG-GTCTTTGC.

Western blot

Western blot was performed as described previously^{40, 43}. Equal amounts of protein were subjected to western blot analysis. Blots were probed with antibodies against Kallistatin (ab187656, Abcam, Cambridge, UK), A β (ab201060, Abcam, Cambridge, UK), Presenilin-1 (ab76083, Abcam, Cambridge, UK), BACE1 (#5606S, Cell Signaling Technology, Boston, USA), APP (#2452S, Cell Signaling Technology, Boston, USA), MAP2 (#4542, Cell Signaling Technology, Boston, USA), PPAR γ (#2435, Cell Signaling Technology, Boston, USA), SP1 (#9389, Cell Signaling Technology, Boston, USA), YY1 (#46395, Cell Signaling Technology, Boston, USA), Notch1 (#3608, Cell Signaling Technology, Boston, USA), Hes1 (#11988, Cell Signaling Technology, Boston, USA), p-tau S202 (ab108387, Abcam, Cambridge, UK), p-tau T231(ab151559, Abcam, Cambridge, UK), p-tau S396 (ab109390, Abcam, Cambridge, UK), tau (ab75714, Abcam, Cambridge, UK), GSK3 β (#70109S, Cell Signaling Technology, Boston, USA), p-GSK3 β Ser9 (#9323, Cell Signaling Technology, Boston, USA), β -actin (A5441-2ml, Sigma, CA, USA), anti-Mouse (#PI200, Vector Laboratories, Burlingame, CA, USA), anti-Rabbit (#PI1000, Vector Laboratories, Burlingame, CA, USA). The signal intensity was quantified using ImageJ (NIH).

Statistical Analysis

The results are expressed as mean $\pm$ SD. Student's *t*-test was applied for comparisons of parametric data between two groups, and one-way ANOVA followed by LSD *t*-test was used to compare differences between more than two different groups (GraphPad Prism software). A *P* value less than 0.05 was considered statistical significance.

Acknowledgements

We are thankful to Zhongshan City People's Hospital, Zhongshan Third People's Hospital, Guangdong Provincial People's Hospital, and Sun Yat-sen Memorial Hospital for kindly providing serum samples for the analyses of this manuscript. We thank Professor Boxing Li for his valuable advice on our study.

List of abbreviations

- A β : amyloid β
- p-tau: hyperphosphorylated tau
- AD: Alzheimer's disease
- T2DM: type 2 diabetes mellitus
- FBG: fasting blood glucose
- TG: triglyceride
- KAL-TG: Kallistatin-transgenic
- WT: wild type mice
- APP: amyloid precursor protein
- BACE1: β -site APP cleaving enzyme 1
- BMI: body mass index
- ICD-10: The International Statistical Classification of Diseases and Related Health Problems 10th Revision
- NINCDS-ADRDA: the National Institute of Neurological and Communicative Disorders and Stroke and the Alzheimer's Disease and Related Disorders Association
- MMSE: mini-Mental State Examination
- KAL-TG-RSG: rosiglitazone-treated group
- KAL-TG-Feno: fenofibrate-treated group
- MWM: morris water maze
- RT-qPCR: real-time quantitative PCR

Declarations

Ethics approval and consent to participate

All patients involved in this study gave their informed consent. This study obtained the institutional review board approval of Medical Ethics of Zhongshan Medical College No. 072 in 2021 and Animal Experiment Ethics of Zhongshan Medical College, Approval No.: SYSU-IACUC-2019-B051.

Consent for publication

Not applicable.

Availability of data and materials

All the data supporting the conclusions of the current study are presented in the figures and they are available from the corresponding authors upon reasonable request. There are no restrictions on data availability. Source data are provided with this paper.

Competing interests

The authors declare that they have no competing interests.

Funding

This study was supported by The National Natural Science Foundation of China (Grants 82070888, 82070882, 82100917, 82273116, 82203661, 81901557, 81870869, and 32071101); Guangdong Natural Science Fund (Grant 2022A1515012423, 2021A1515010434, 2023A1515010316, and 2023A1515010214); Key Sci-Tech Research Project of Guangzhou Municipality, China (Grants

202201010820, 202102020955); Key Project of Nature Science Foundation of Guangdong Province, China (Grant 2019B1515120077); National Key R&D Program of China (Grant 2018YFA0800403); Guangdong Special Support Program for Young Top Scientist (Grant 201629046); China Postdoctoral Science Foundation (Grant 2021M703679, 2022M713594, BX20220360); Health Commission of Guangdong Province (Grants A2022161) ; Guangzhou Municipal Science and Technology Bureau (Grants SL2022A04J01575); Guangdong Key Project in “Development of new tools for diagnosis and treatment of Autism” (2018B030335001), Research and Development Plan of Key Areas of Guangzhou Science and Technology Bureau (2020070030001), Open Research Funds of State Key Laboratory of Ophthalmology (2020KF03); Guangzhou Key Laboratory for Metabolic Diseases(202102100004); The Science and Technology Planning Project of Guangdong Province 2023B1212060018.

Authors' contributions

G. Gao, X Yang, B Jiang, and P Jiang were involved in the concept and design of the study. W. Qi, Y. Long, Z. Li, Z. Zhao J. Shi, D Zhu, Z Zhao, W. Xie, L. Wang, T Zhou were responsible for conducting the experiments. W. Qi, Y. Long and T Zhou drafted the manuscript and G. Gao revised the manuscript. P Jiang, Y. Long and Z. Li were responsible for data analysis. All authors contributed to the interpretation of data and provided revisions to the manuscript. G. Gao will act as guarantor for the study. All authors read and approved the final manuscript. W. Qi, Y. Long and Z. Li contributed equally to this study.

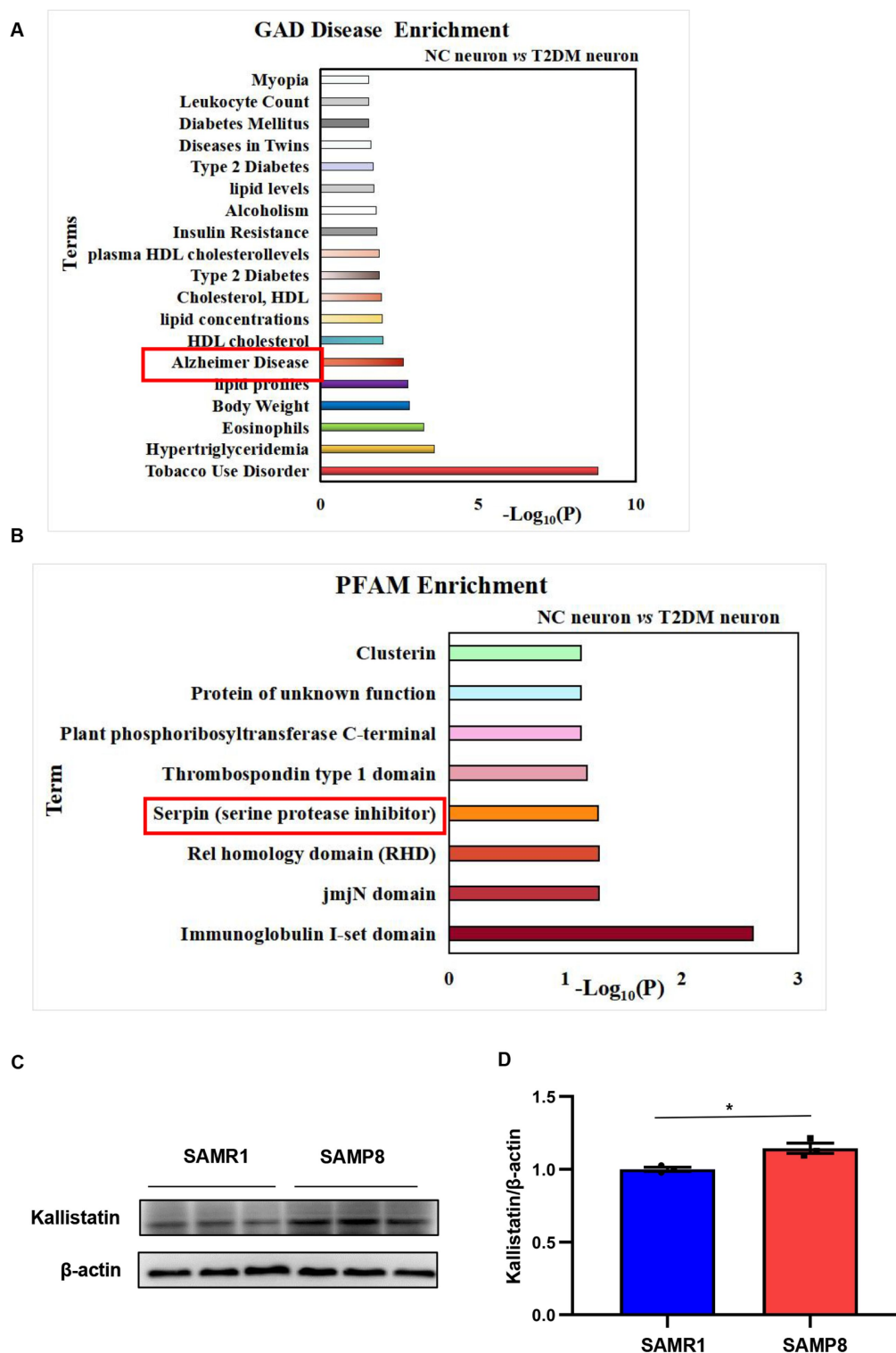


Fig. S1

(A-B) GAD disease enrichment analysis (A) and PFAM analysis (B) result. Differentially expressed genes in neurons were obtained from GSE161355, and GAD disease enrichment was analyzed on David database. (C-D) Western blot analysis of Kallistatin expression in aging model SAMP8 and corresponding control SAMR1 mice hippocampal tissue samples, then statistically analyzing the above results. β -Actin served as a loading control. Error bars represent the standard deviation (SD); three asterisks, $p < 0.001$.

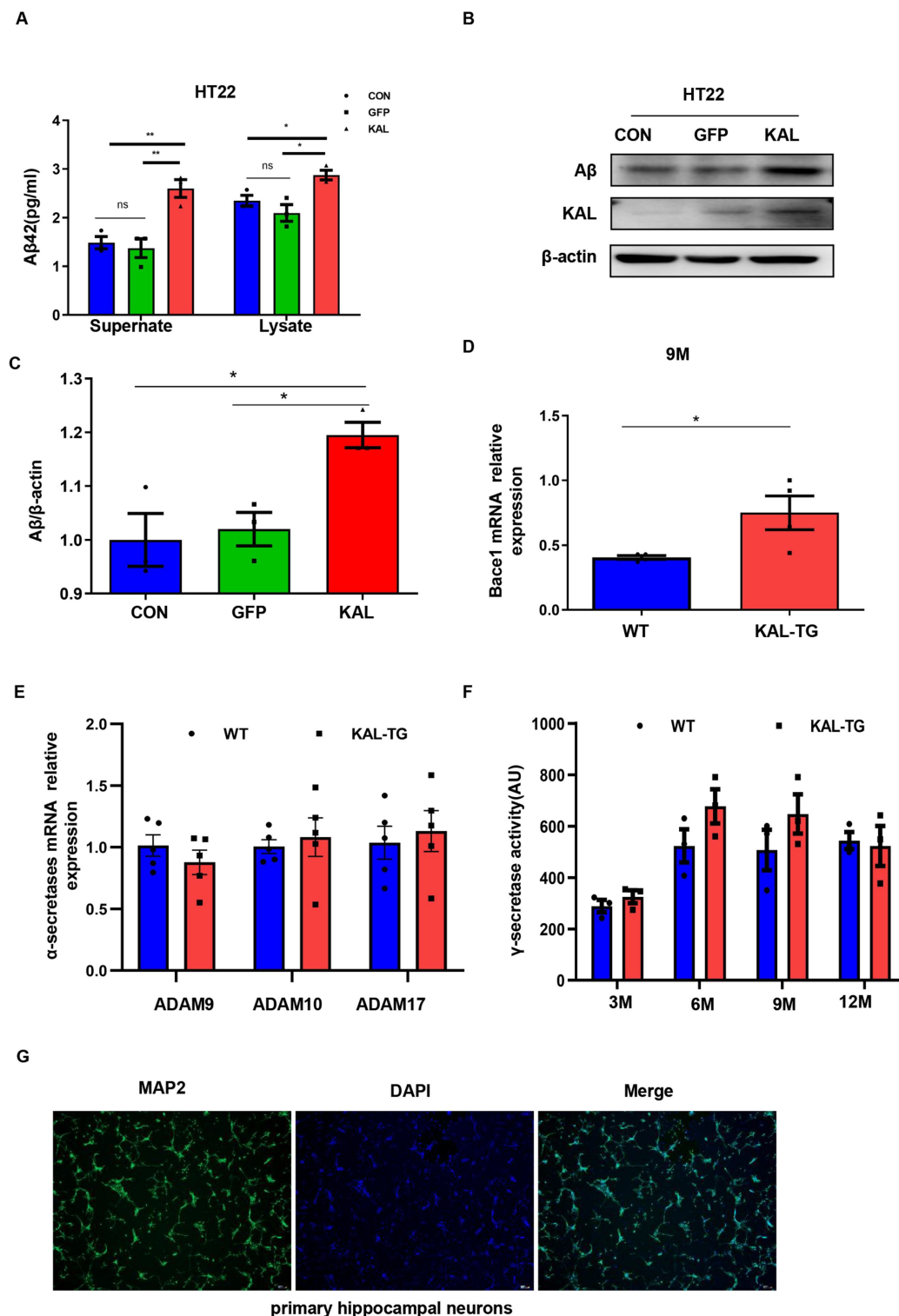


Fig. S2

(A) HT22 cells were infected with adenovirus to overexpress Kallistatin for 48h. A β 42 concentration of supernate and cell lysate was quantified by ELISA. (B-C) Western blot analysis of A β protein level in HT22 cells infected with overexpressing Kallistatin adenovirus and control groups for 48h, then statistical analysis of Kallistatin protein levels. (D) BACE1 mRNA expression in hippocampal tissue. (E) α -secretases (ADAM9, ADAM10, ADAM17) mRNA expression in hippocampal tissue. (F) γ -secretase activity of each group's hippocampal tissue was measured by ELISA. (G) Primary hippocampal neurons were identified with MAP2. Error bars represent the standard deviation (SD), one asterisk, $p < 0.05$, two asterisks, $p < 0.01$.

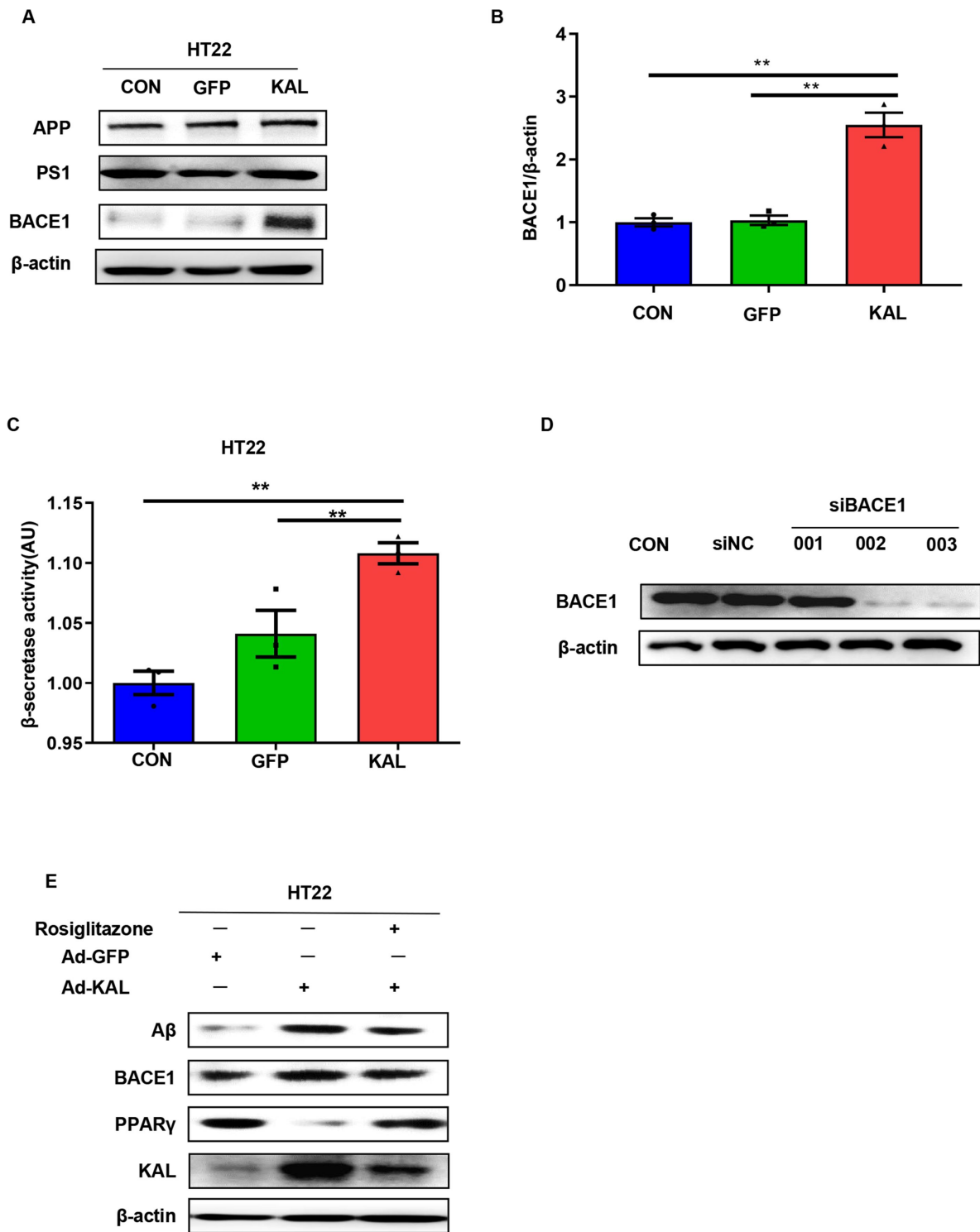


Fig. S3

(A) The relevant protein levels in HT22 cells infected with overexpressing Kallistatin adenovirus during A β generation were determined by western blot analysis. (B) Statistical analysis of BACE1 expression in HT22 cells. (C) β -secretase activity of HT22 cells infected with overexpressing Kallistatin adenovirus and control adenovirus was measured by ELISA. (D) Primary hippocampal neurons were infected with BACE1 siRNA for 72h. Western blot analysis of BACE1 protein levels. (E) HT22 cells were treated with PPAR γ agonist rosiglitazone (10nM) for 12h, then infected with adenovirus to overexpress Kallistatin for 48h. Western blot analysis of A β and BACE1 protein levels. β -actin served as a loading control. Error bars represent the standard deviation (SD), two asterisks, $p < 0.01$.

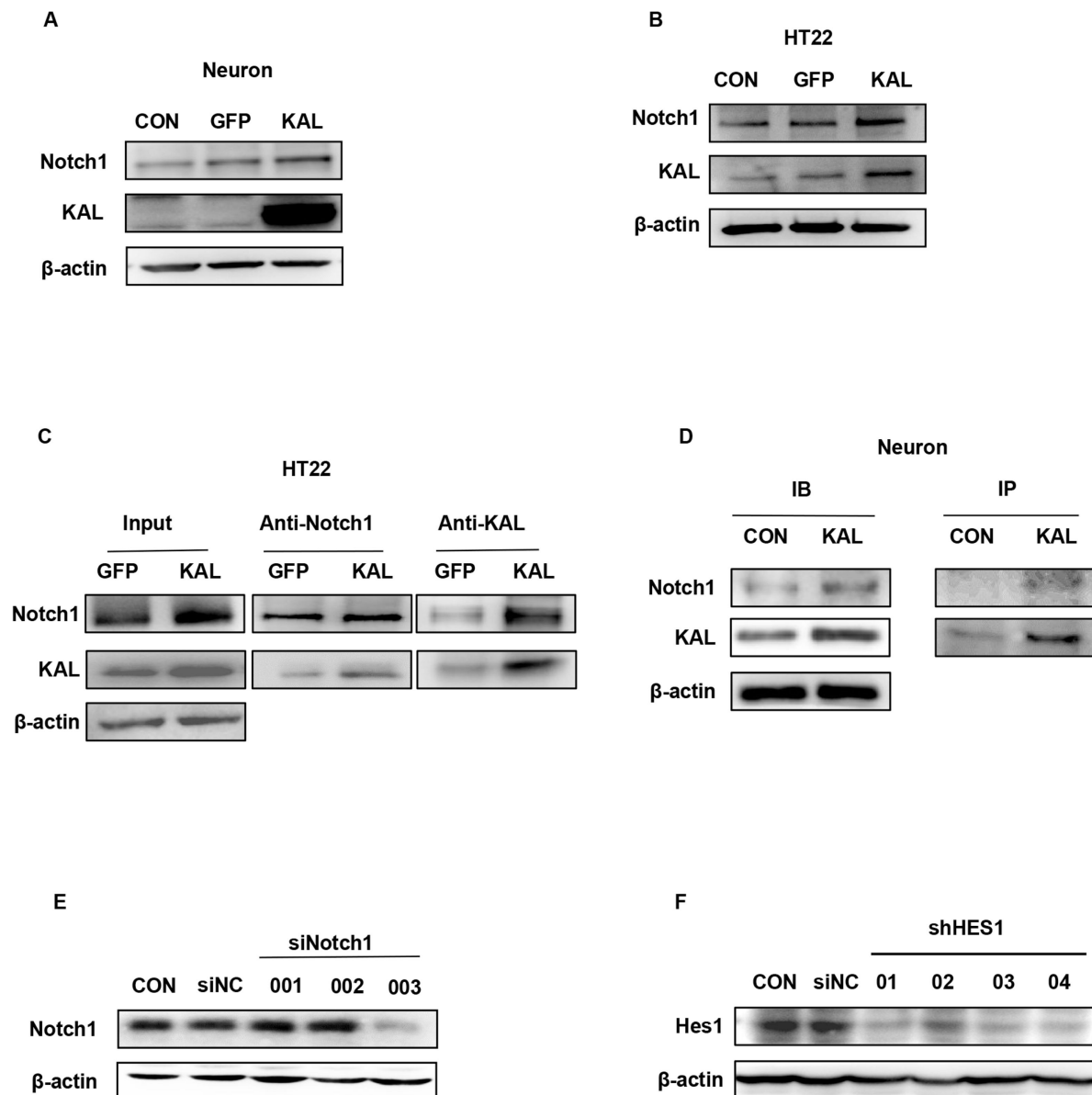


Fig. S4

(A-B) Western blot analysis of Notch1 protein level in primary hippocampal neurons and HT22 cells infected with overexpressing Kallistatin adenovirus and control groups. (C) HT22 cells were infected with overexpressing Kallistatin adenovirus for 48h, and Co-IP analysis was conducted to verify whether Kallistatin can bind to the Notch1 receptor. β -actin served as a loading control. (D) Primary hippocampal neurons were treated with Kallistatin protein for 72h, then IP analysis. (E) Primary hippocampal neurons were infected with Notch1 siRNA for 72h. Western blot analysis of BACE1 protein levels. (F) HT22 cells were infected with HES1 shRNA for 48h. Western blot analysis of HES1 protein levels.

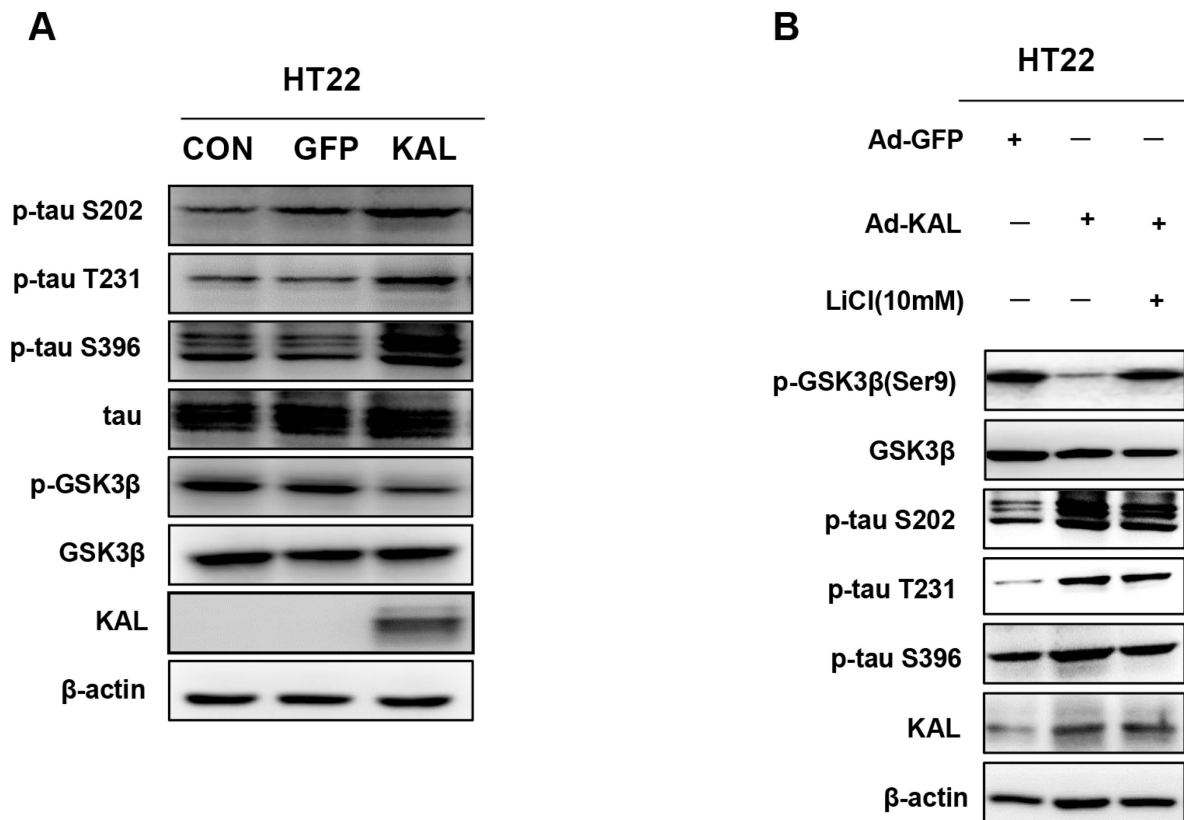


Fig. S5

(A) Western blot analysis of GSK-3β, p-GSK-3β, tau, p-tau(Ser9, T231, S396) in HT22 cells infected with overexpressing Kallistatin adenovirus and control groups. (B) Western blot analysis of GSK-3β, p-GSK-3β, p-tau(Ser9, T231, S396) in HT22 cells infected with overexpressing Kallistatin adenovirus and control groups for 24h, then treated with LiCl(10 mM) for 24h.

	NC	AD	<i>p value</i> (Dementia vs NC)
N	61	56	NA
Age	49-82	52-85	****
Average age	64±8.53	73.02±9.43	NA
Male	29	25	NS
Female	32	31	
GLU (mmol/L)	5.07±0.65	6.73±2.81	****
TC (mmol/L)	4.63±0.58	4.50±1.12	NS
TG (mmol/L)	1.09±0.50	1.41±0.65	**

N is an abbreviation for number; GLU is an abbreviation for glucose; TC is an abbreviation for total cholesterol; TG is an abbreviation for triglyceride; NA indicates not available; NS indicates no significance. two asterisk, $p < 0.01$, four asterisk, $p < 0.0001$.

Table S1

Clinical characteristic of AD patients

Table S2

Clinical characteristic of AD patients with DM

	NC	AD + DM	<i>p value</i> (<i>Dementia+DM</i> vs <i>NC</i>)
N	61	26	NA
Age	49-82	52-92	****
Average age	64±8.53	76.77±8.69	NA
Male	29	15	NS
Female	32	11	NS
GLU (mmol/L)	5.07±0.65	8.12±3.05	****
TC (mmol/L)	4.63±0.58	4.29±1.04	NS
TG (mmol/L)	1.08±0.50	1.69±0.68	****

N is an abbreviation for number; GLU is an abbreviation for glucose; TC is an abbreviation for total cholesterol; TG is an abbreviation for triglyceride; NA indicates not available; NS indicates no significance. four asterisk, $p < 0.0001$.

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Reviewer #1 (Public Review):

Summary:

Qi and colleagues investigated the role of the Kallistatin pathway in increasing hippocampal amyloid-β plaque accumulation and tau hyperphosphorylation in Alzheimer's disease, linking the increased Kallistatin level in diabetic patients with a higher risk of Alzheimer's disease development. A Kallistatin-overexpressing animal model was utilized, and memory impairment was assessed using Morris water maze and Y-maze. Kallistatin-related pathway protein levels were measured in the hippocampus, and phenotypes were rescued using fenofibrate and rosiglitazone. The current study provides evidence of a novel molecular mechanism linking diabetes and Alzheimer's disease and suggests the potential use of fenofibrate to alleviate memory impairment. However, several issues need to be addressed before further consideration.

Strengths:

The findings of this study are novel. The findings will have great impacts on diabetes and AD research. The studies were well conducted, and the results were convincing.

Weaknesses:

- (1) The mechanism by which fenofibrate rescues memory loss in Kallistatin-transgenic mice is unclear. As a PPARalpha agonist, does fenofibrate target the Kallistatin pathway directly or indirectly? Please provide a discussion based on literature supporting either possibility.
- (2) The current study exclusively investigated the hippocampus. What about other cognitive memory-related regions, such as the prefrontal cortex? Including data from these regions or discussing the possibility of their involvement could provide a more comprehensive understanding of the role of Kallistatin in memory impairment.
- (3) Fenofibrate rescued phenotypes in Kallistatin-transgenic mice while rosiglitazone, a PPARgamma agonist, did not. This result contradicts the manuscript's emphasis on a PPARgamma-associated mechanism. Please address this inconsistency.
- (4) Most of the immunohistochemistry images are unclear. Inserts have similar magnification to the original representative images, making judgments difficult. Please provide larger inserts with higher resolution.
- (5) The immunohistochemistry images in different figures were taken from different hippocampal subregions with different magnifications. Please maintain consistency, or explain why CA1, CA3, or DG was analyzed in each experiment.
- (6) Figure 5B is missing a title. Please add a title to maintain consistency with other graphs.
- (7) Please list statistical methods used in the figure legends, such as t-test or One-way ANOVA with post-hoc tests.

<https://doi.org/10.7554/eLife.99462.1.sa2>

Reviewer #2 (Public Review):

Summary:

The study links Alzheimer's disease (AD) with metabolic disorders through elevated Kallistatin levels in AD patients. Kallistatin-overexpressing mice show cognitive decline, increased A β and tau pathology, and impaired hippocampal function. Mechanistically, Kallistatin enhances A β production via Notch1 and promotes tau phosphorylation through GSK-3 β activation. Fenofibrate improves cognitive deficits by reducing A β and tau phosphorylation in these mice, suggesting therapeutic potential in AD linked to metabolic syndromes.

Strengths:

This study presents novel insights into AD pathogenesis and provides strong evidence about the mechanistic roles of Kallistatin, and the therapeutic potential of fenofibrate in AD.

Weaknesses:

It was suggested that Kallistatin is primarily produced by the liver. The study demonstrates increased Kallistatin levels in the hippocampus tissue of AD mice. It would be valuable to clarify if Kallistatin is also increased in the liver of AD mice, providing a comprehensive understanding of its distribution in disease states.

Does Kallistatin interact directly with Notch1 ligands? Clarifying this interaction mechanism would enhance understanding of how Kallistatin influences Notch1 signaling in AD pathology.

Is there any observed difference in AD phenotype between male and female Kallistatin-transgenic (KAL-TG) mice? Including this information would address potential gender-specific effects on cognitive decline and pathology.

It is recommended to include molecular size markers in Western blots for clarity and accuracy in protein size determination.

The language should be revised for enhanced readability and clarity, ensuring that complex scientific concepts are communicated effectively to a broader audience.

<https://doi.org/10.7554/eLife.99462.1.sa1>

Reviewer #3 (Public Review):

Summary:

The authors investigated the role of kallistatin in metabolic abnormalities associated with AD. They found that Kallistatin promotes A β production by binding to the Notch1 receptor and upregulating BACE1 expression. They identified that Kallistatin is a key player that mediates A β accumulation and tau hyperphosphorylation in AD.

Strengths:

This manuscript not only provides novel insights into the pathogenesis of AD, but also indicates that the hypolipidemic drug fenofibrate attenuates AD-like pathology in Kallistatin transgenic mice.

Weaknesses:

The authors did not illustrate whether the protective effect of fenofibrate against AD depends on kallistatin.

The conclusions are supported by the results, but the quality of some results should be improved.

<https://doi.org/10.7554/eLife.99462.1.sa0>